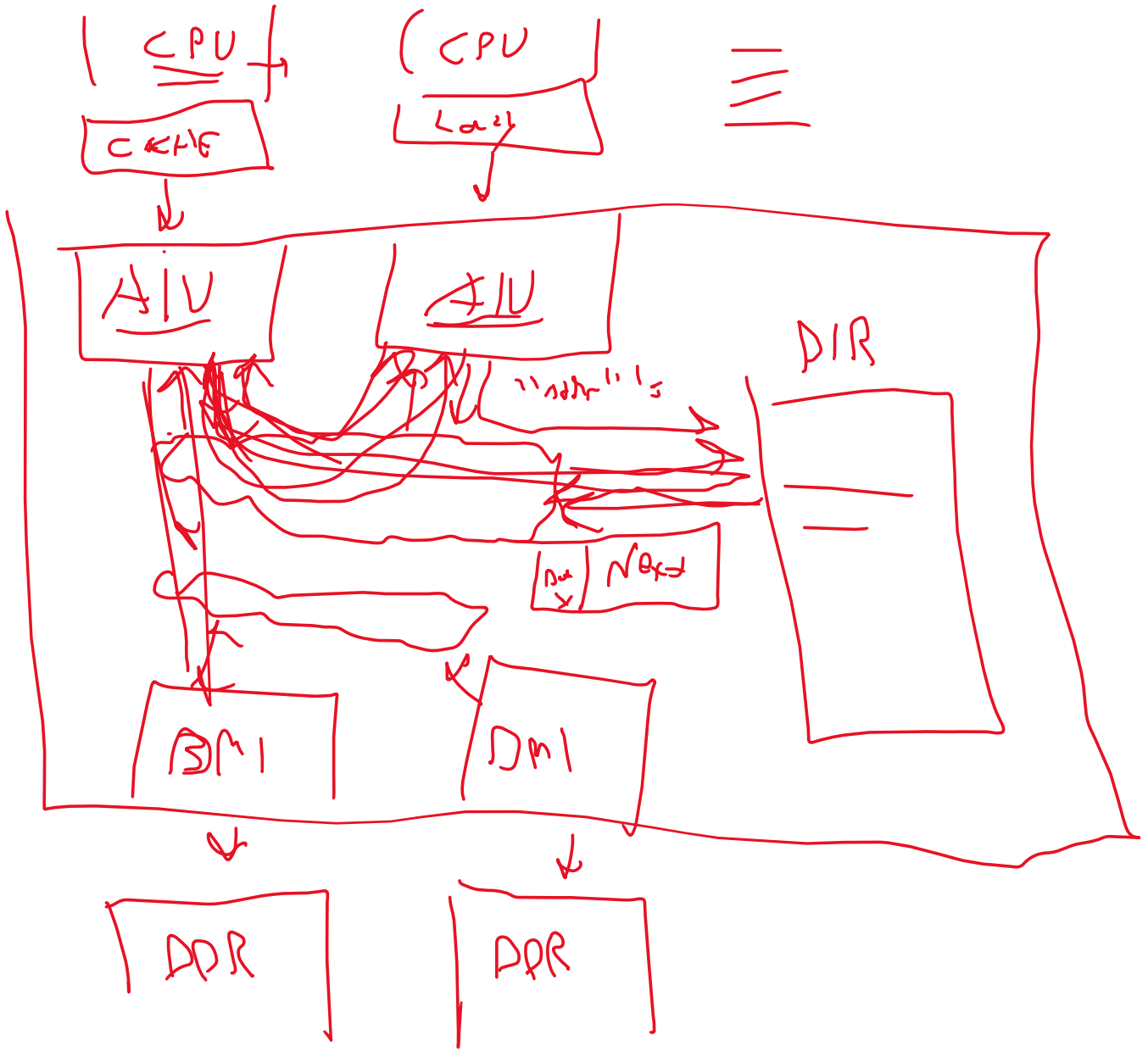
SoC Interconnect Design

Eric Taylor – Co-Founder & Chief HW Architect @ Baud

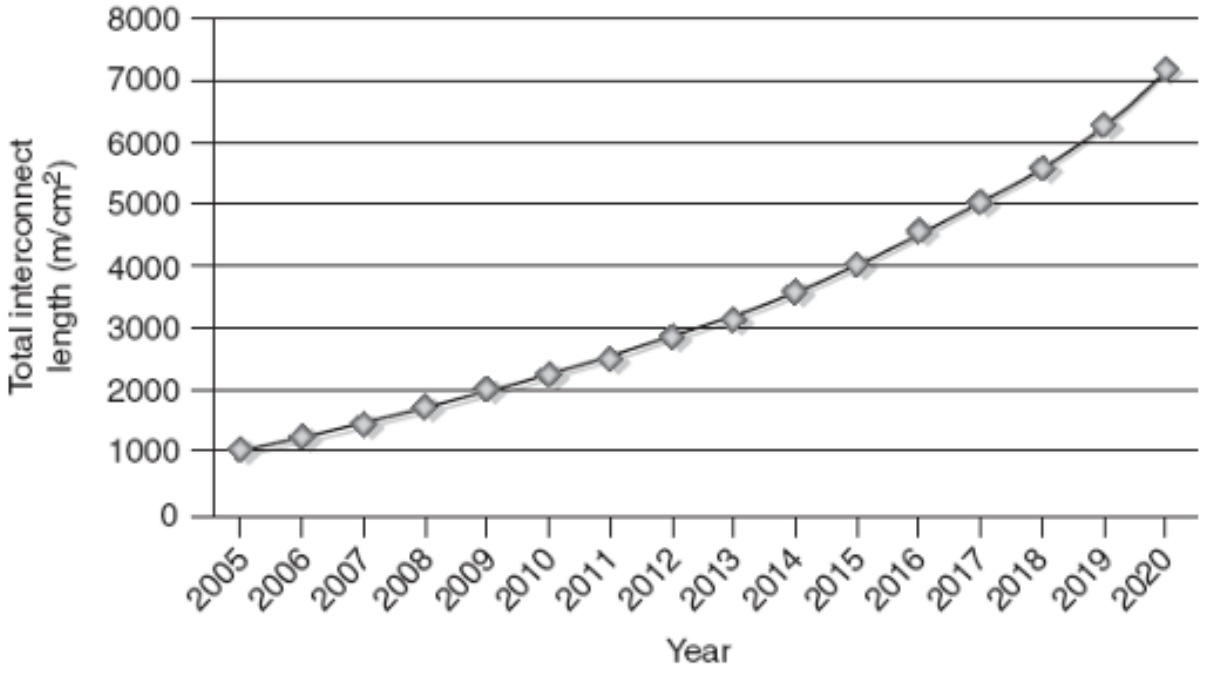
Outline

- Background Concepts
  - General NoC Concepts
  - Topology
  - Routing
  - Packets
  - Switching
  - Arbitration
  - Flow Control
  - QoS
- Router Design Example

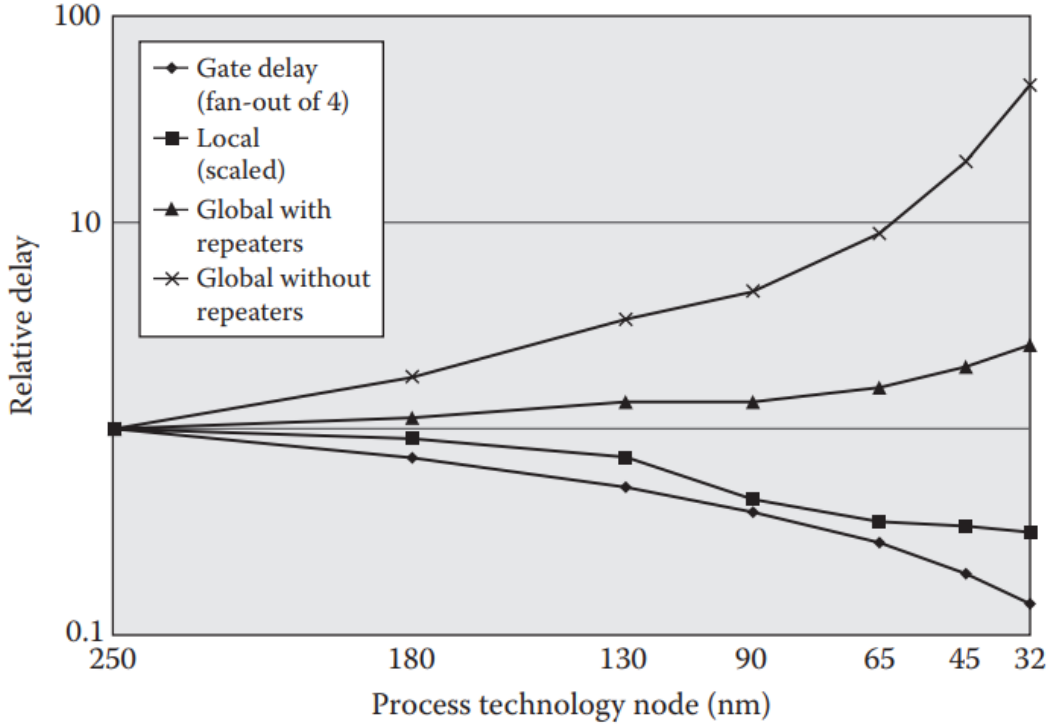


On-Chip Communication Trends

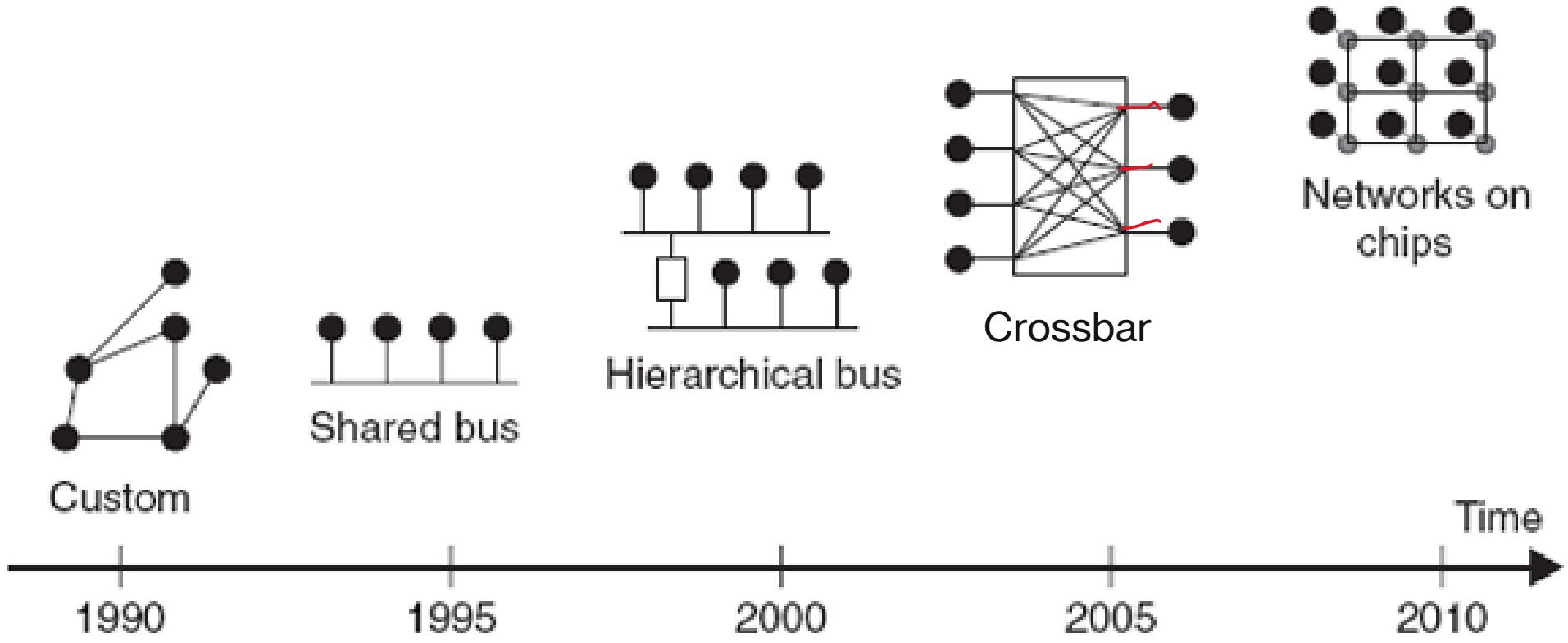
More Wires are Used



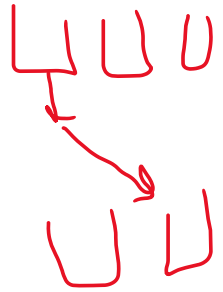
Cost of Distance Changes Over Time



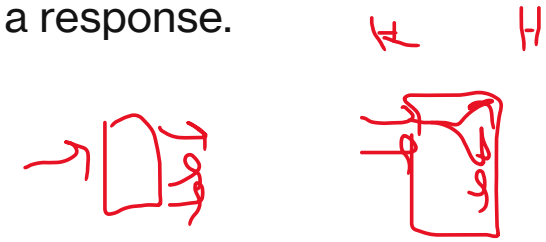
Trends in Wire / Performance Trade-Off



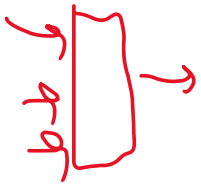
Basic Interconnect Terms



- **Initiator / Target (Master / Slave)**
  - The initiator starts a transaction. The target accepts the request and returns a response.
- **Router**
  - Forwards packets through output links toward their destinations.



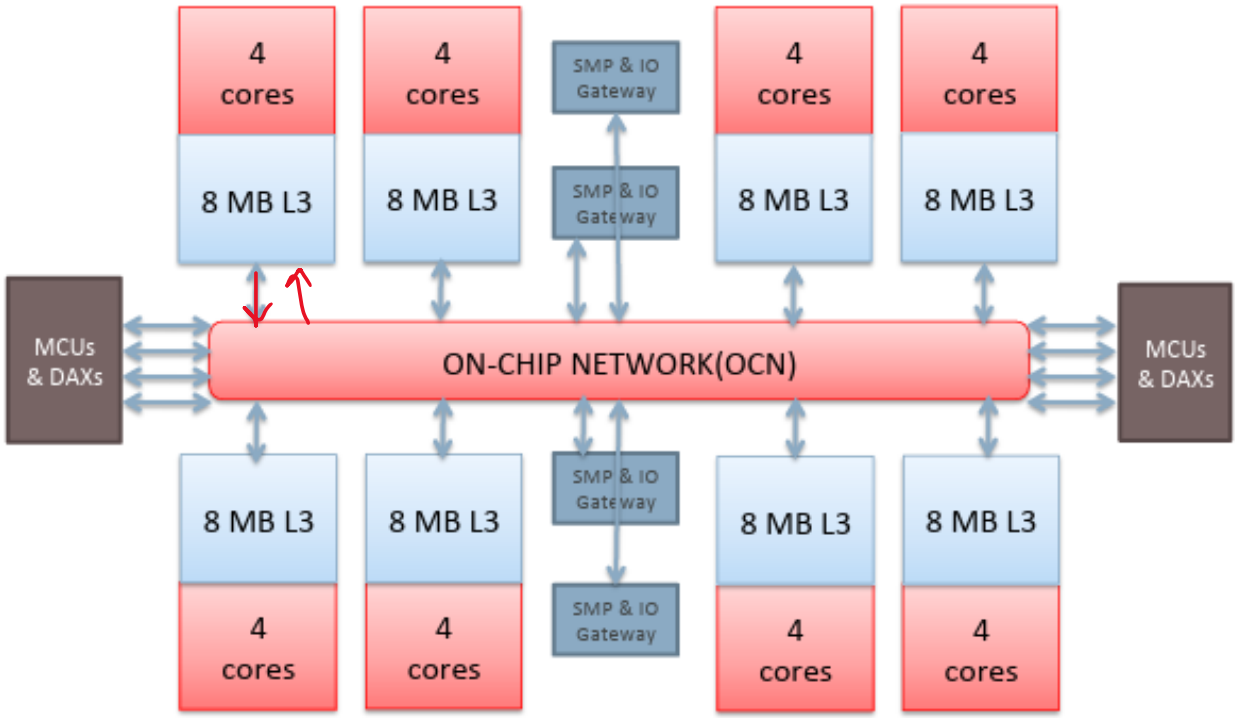
- **Arbiter**
  - Selects which requester gets a shared resource when requests compete.



- **Gasket / Adapter**
  - Connects an IP interface to the fabric, adapting protocol or packet format as needed.
- **Upstream / Downstream**
  - Toward the initiator / toward the target, respectively, for the whole transaction.

Network On Chip

Processor Block Diagram



- An On-Chip Network (OCN) that connects
  - 8 partitions
  - 4 SMP& IO Gateways
  - 4 scheduler and 4 DAX instances on each side
- 64GBps per OCN port per direction
- Low latency cross-partition access

ORACLE

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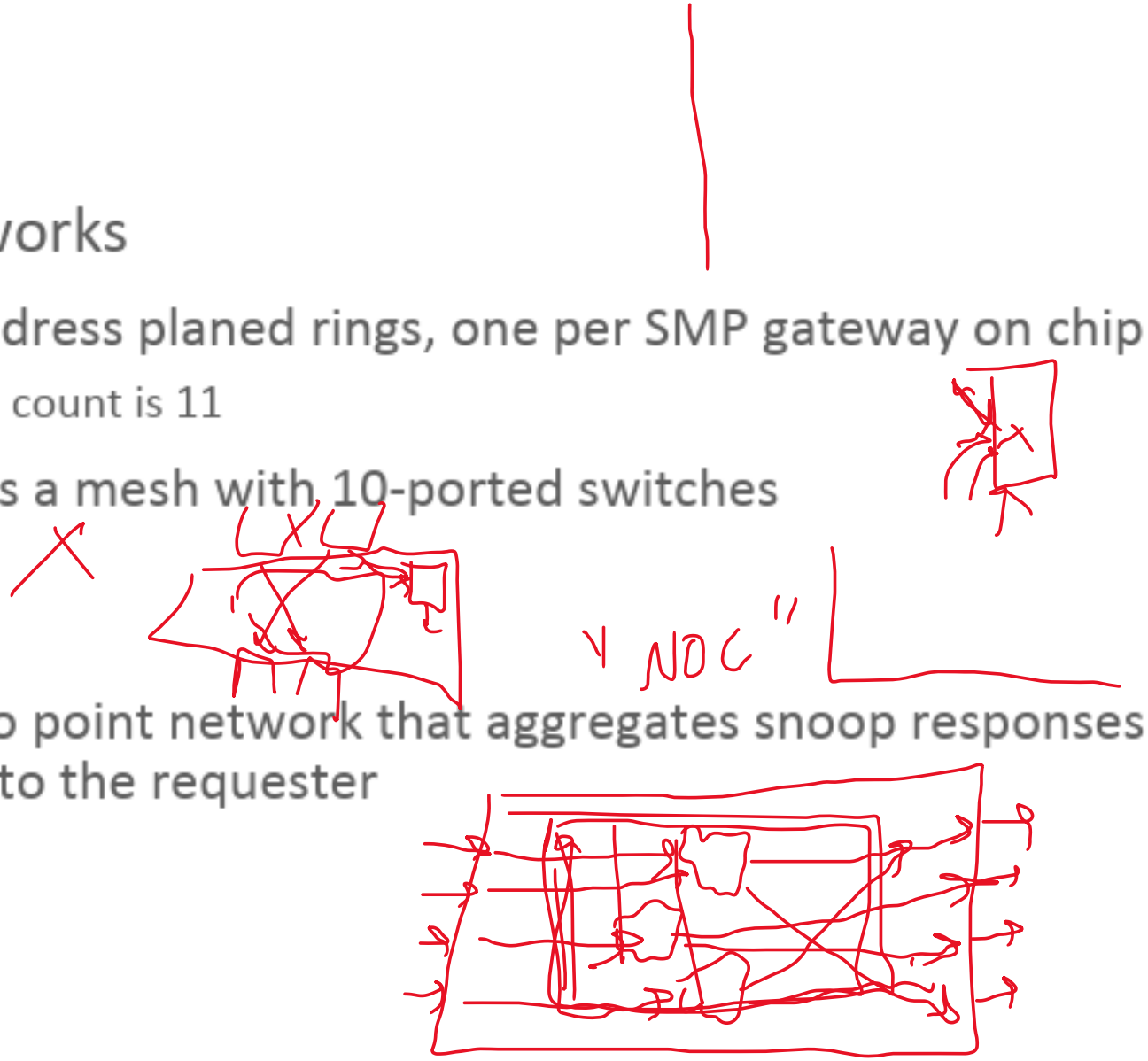
Interconnect Metrics

- **Bandwidth / Throughput**
  - Peak capacity versus sustained useful data delivered per second. State the link or endpoint being measured.
- **Concurrency**
  - Independent transfers at the same time, and transactions outstanding before completion.
- **Latency**
  - Time from request to first response or full completion. Measure both idle and loaded latency.
- **Quality of Service (QoS)**
  - Per-client bandwidth, latency, and fairness when traffic competes. A priority tag alone gives no guarantee.

On-Chip Network(OCN)

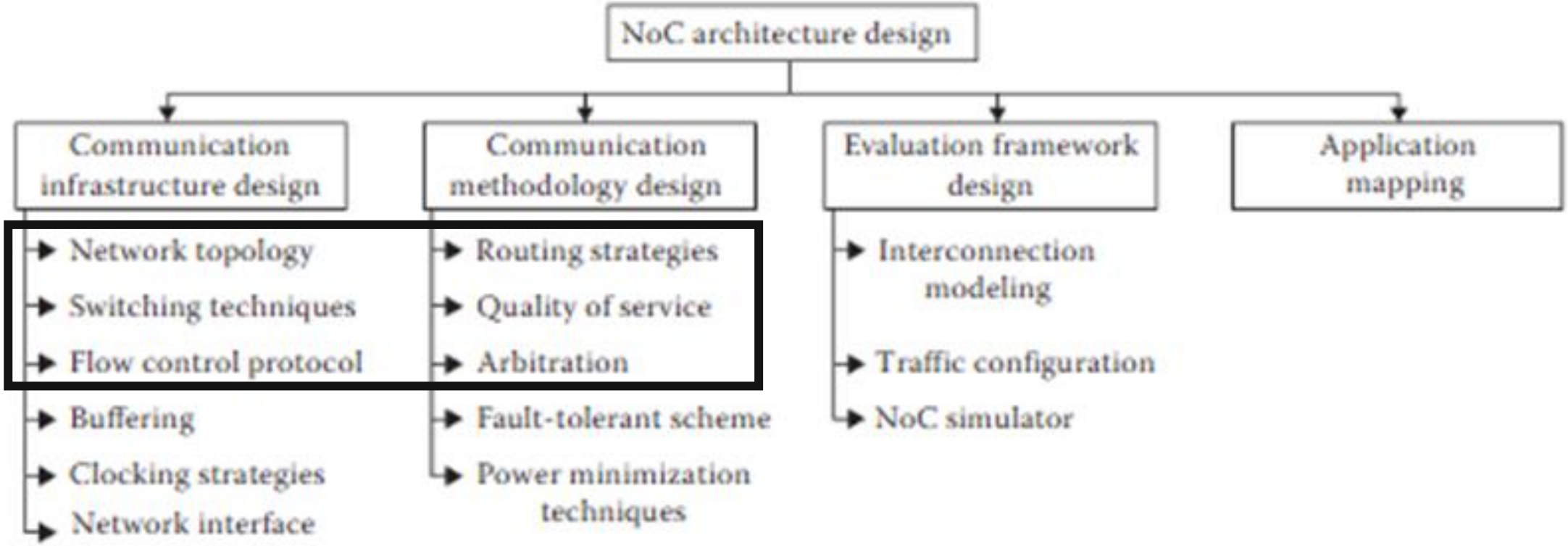
The OCN consists of three networks

- Requests are broadcast on four address planed rings, one per SMP gateway on chip
  - No queuing on the ring, maximum hop count is 11
- The Data network is constructed as a mesh with 10-ported switches
  - Provides ~512GBps per cross-section
  - Unloaded latency <6ns
- The Response network is a point to point network that aggregates snoop responses from all partitions before delivery to the requester



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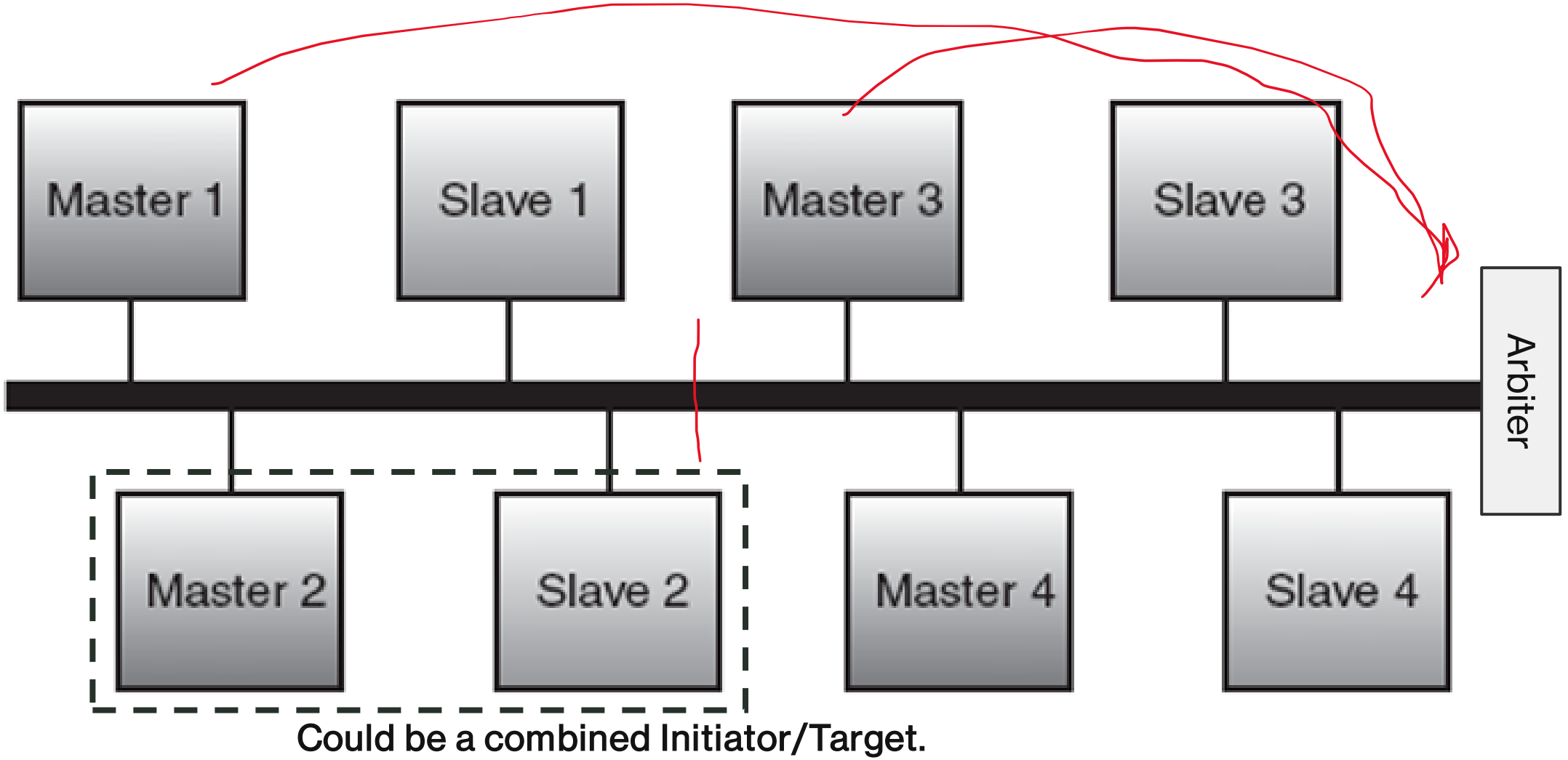
What goes into designing a NoC?



Topology Concepts

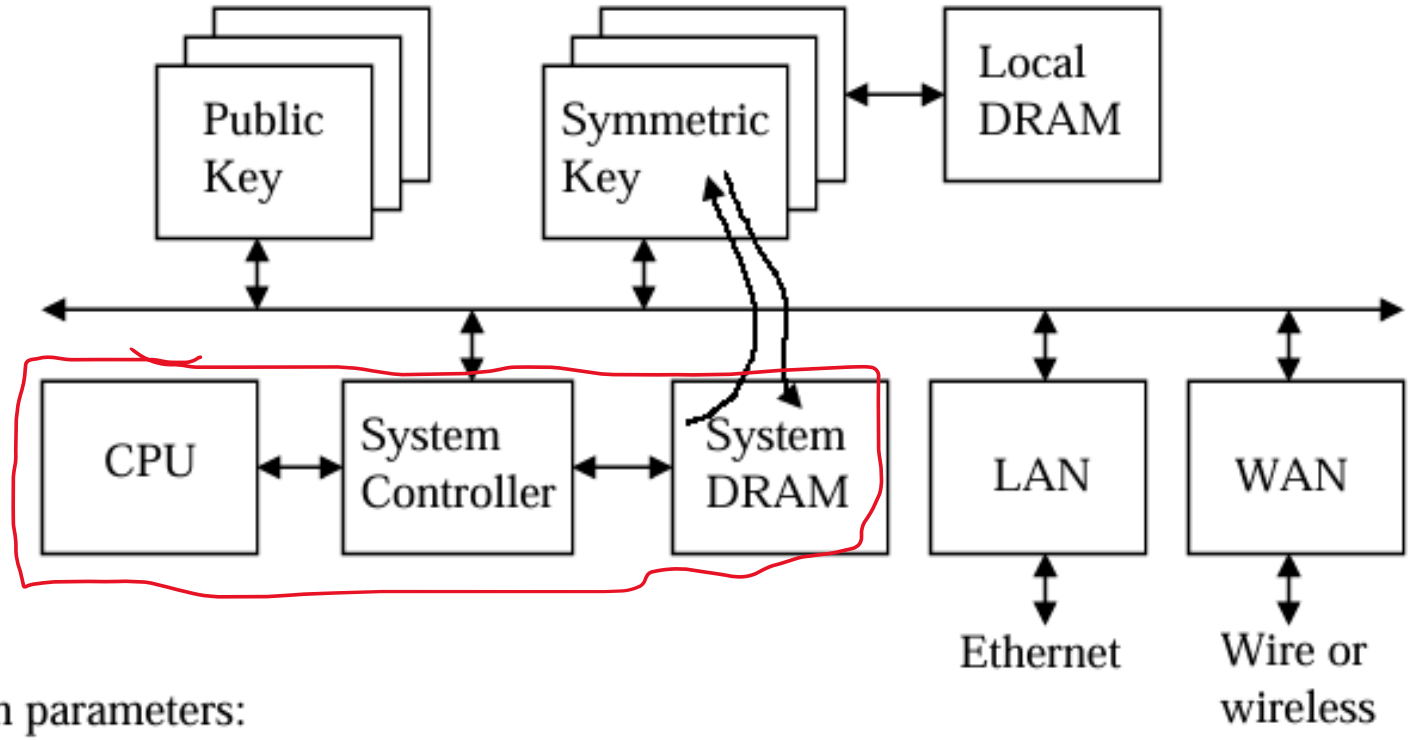
- Bus
- Crossbar
- Ring
- Mesh
- Custom

Bus Topology



Bus (1998)

Typical Hardware System

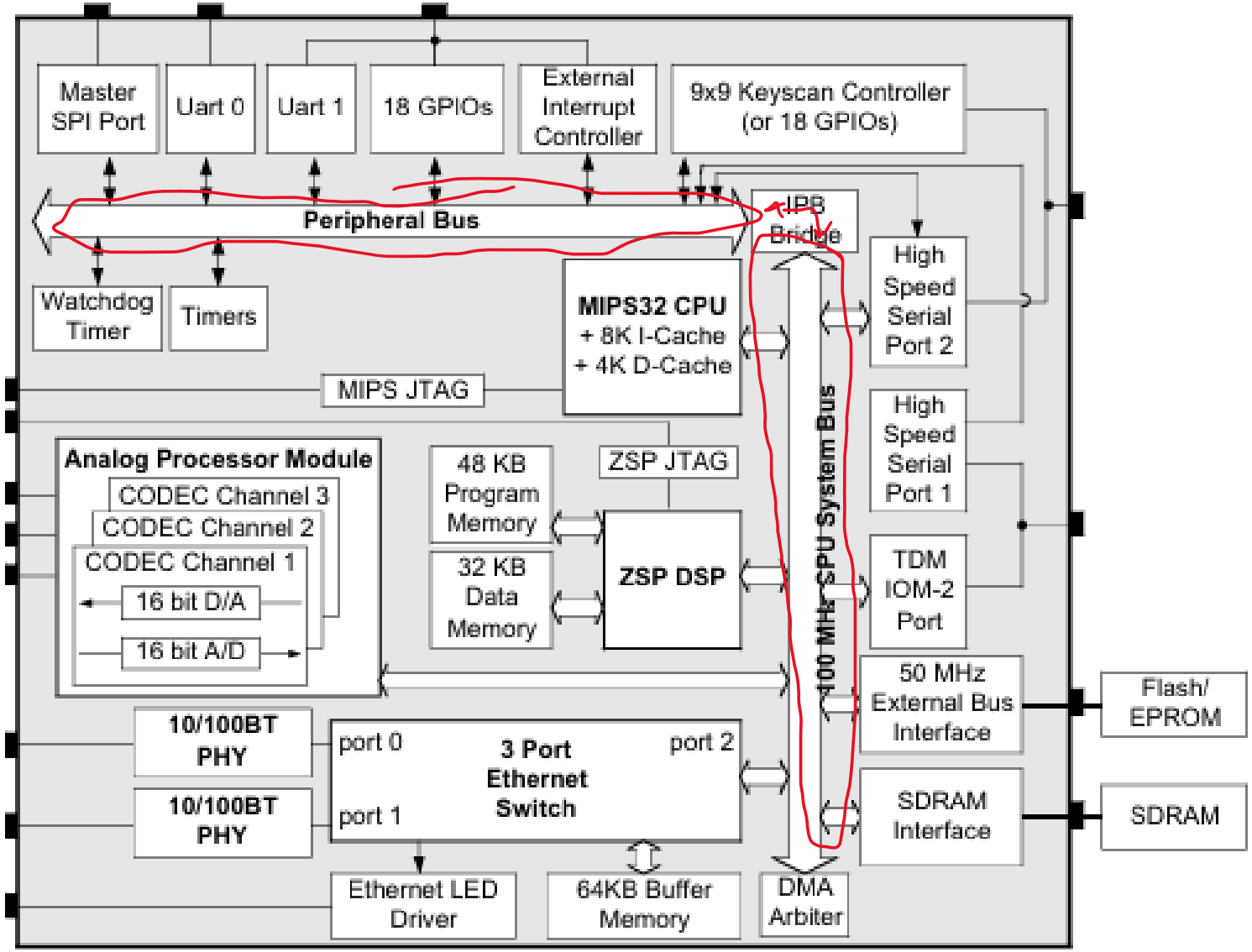


System parameters:

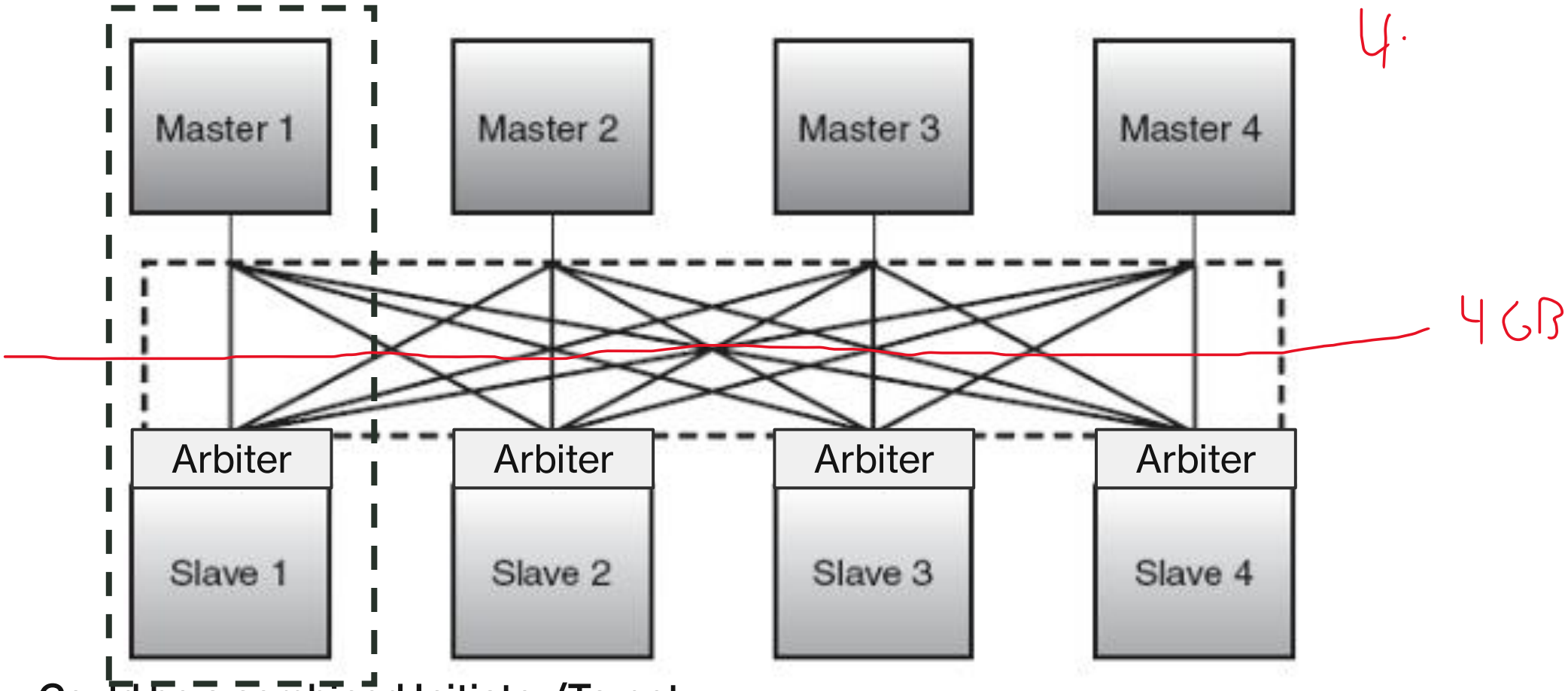
- All-in-one vs. separate public key and symmetric key silicon/performance scalability
- DMA capabilities and bus bandwidth
- Dedicated vs. shared RAM

hifn Network Security Processors

Bus

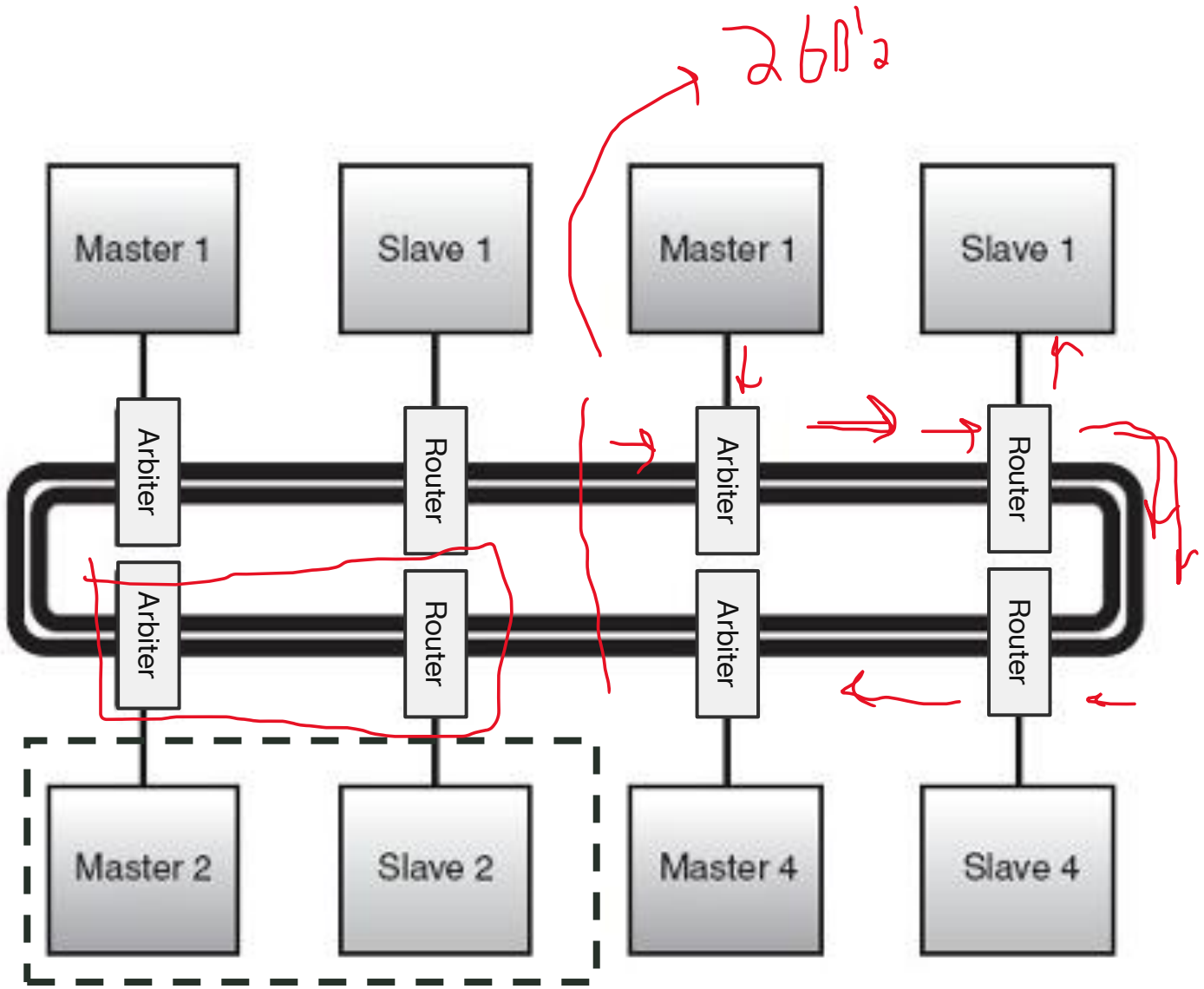

Master SPI Port
Uart 0
Uart 1
18 GPIOs
External Interrupt Controller
9x9 Keyscan Controller (or 18 GPIOs)
Peripheral Bus
IPB Bridge
Watchdog Timer
Timers
MIPS32 CPU
+ 8K I-Cache
+ 4K D-Cache
High Speed Serial Port 2
MIPS JTAG
High Speed Serial Port 1
Analog Processor Module
CODEC Channel 3
CODEC Channel 2
CODEC Channel 1
16 bit D/A
16 bit A/D
48 KB Program Memory
32 KB Data Memory
ZSP JTAG
ZSP DSP
TDM IOM-2 Port
100 MHz CPU System Bus
50 MHz External Bus Interface
Flash/ EPROM
10/100BT PHY
10/100BT PHY
port 0
port 1
3 Port Ethernet Switch
port 2
SDRAM Interface
SDRAM
Ethernet LED Driver
64KB Buffer Memory
DMA Arbiter

Crossbar Topology



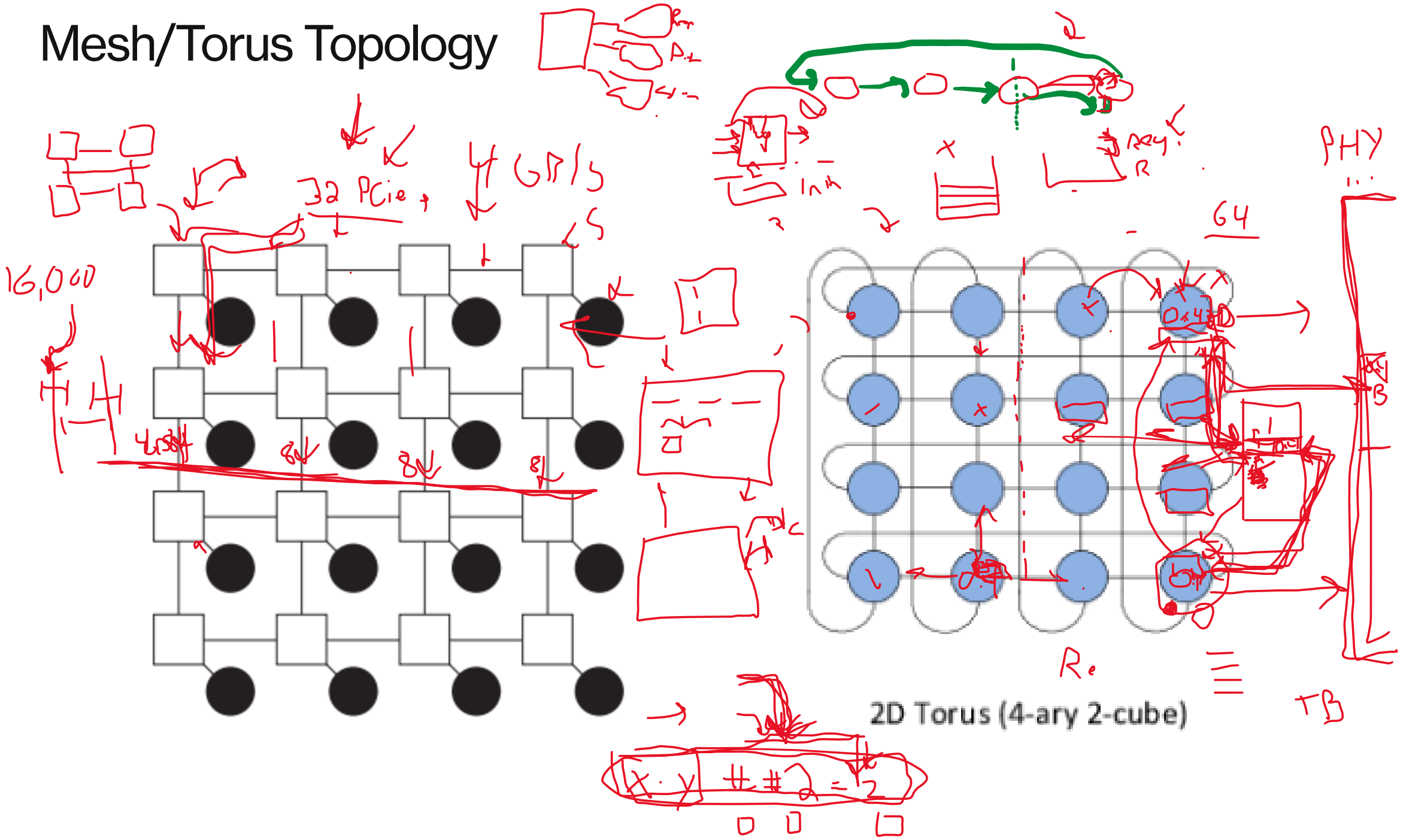
Could be a combined Initiator/Target.

Ring Topology

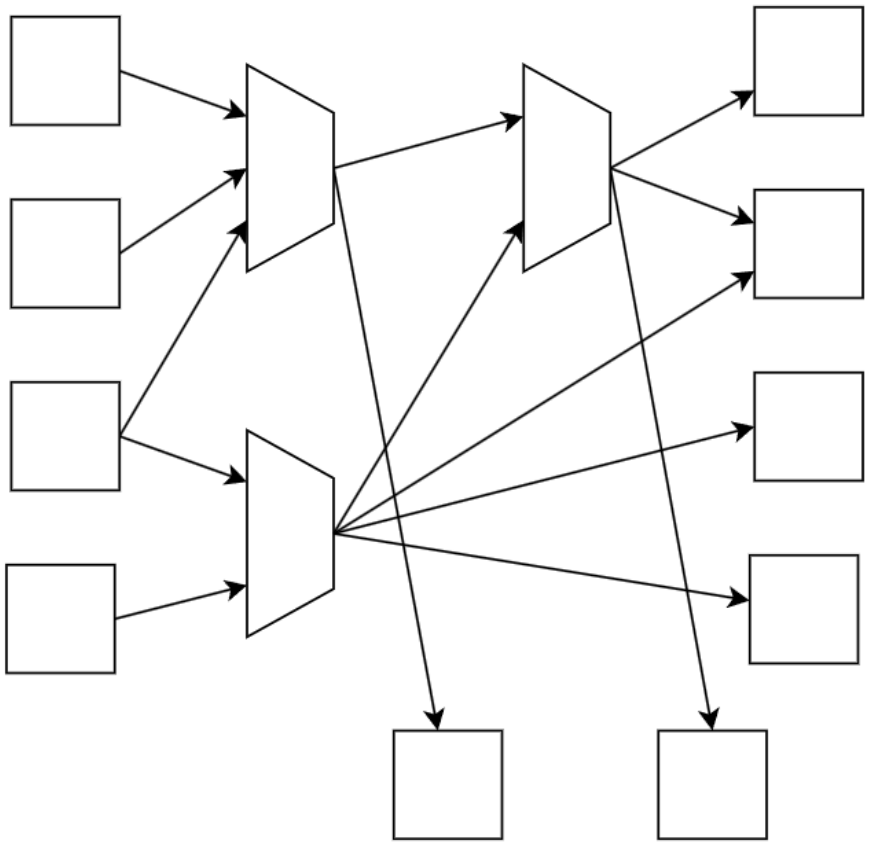


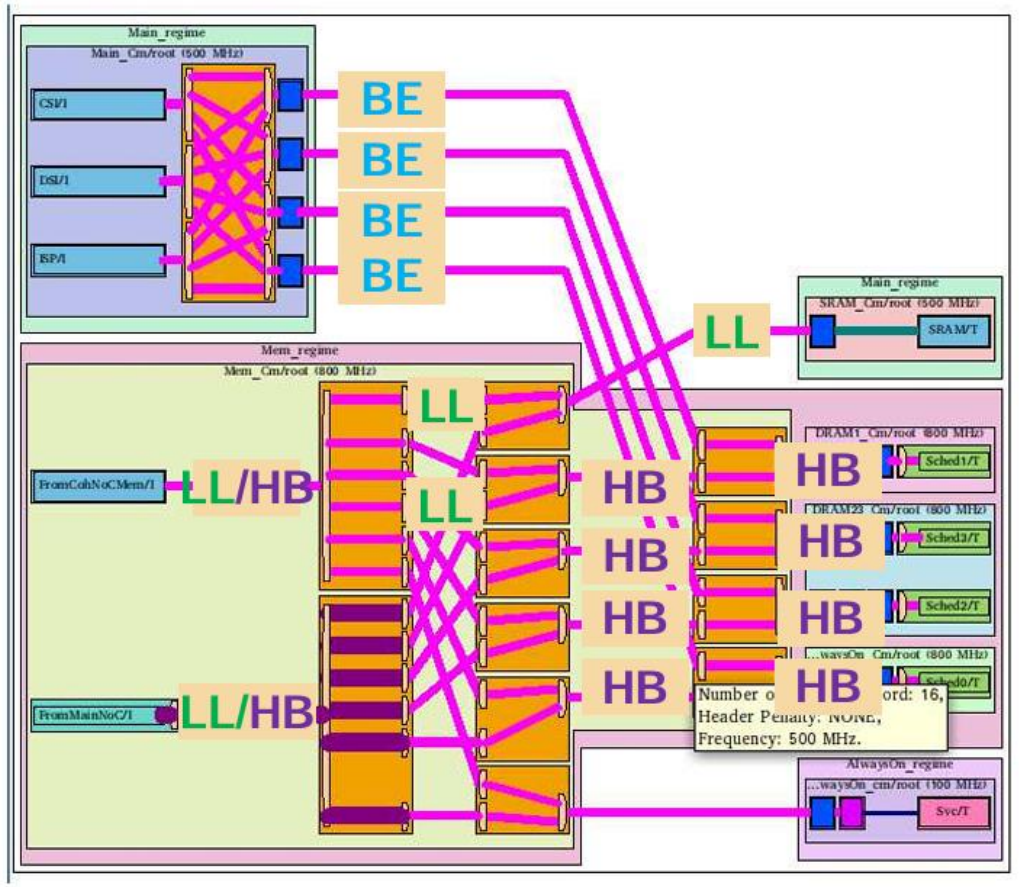
Could be a combined Initiator/Target.

Mesh/Torus Topology



Irregular Topologies





Performance Cost Tradeoff

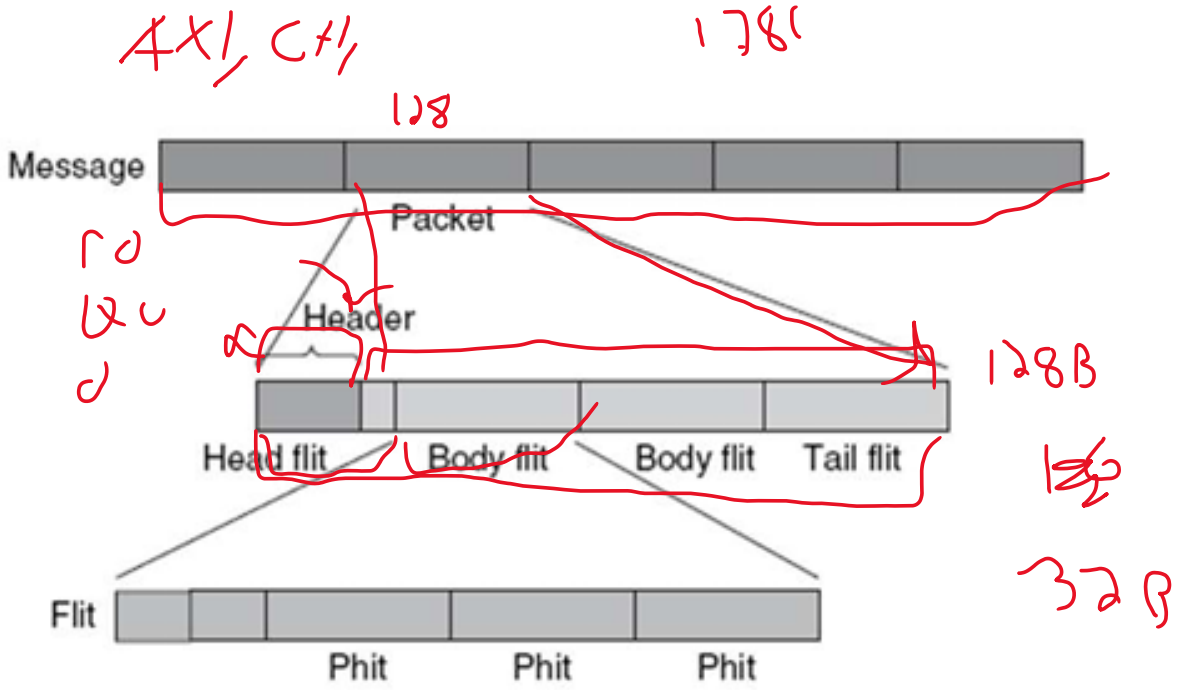
	Bus	Ring	Double Ring	Mesh (NxN)	Crossbar
Performance					
(Bisection BW)	1	2	4	sqrt(N)	N*N
(WC Latency)	1	N	N/2	2*(sqrt(N)-1)	1
Cost					
(Ports/Switch)	1 Overall	3	5	5*	N
(Total Links)	1	N	2*N	2*sqrt(N)*(sqrt(N)-1)	N*N

Packetization Concepts

- Packets
- Packetization

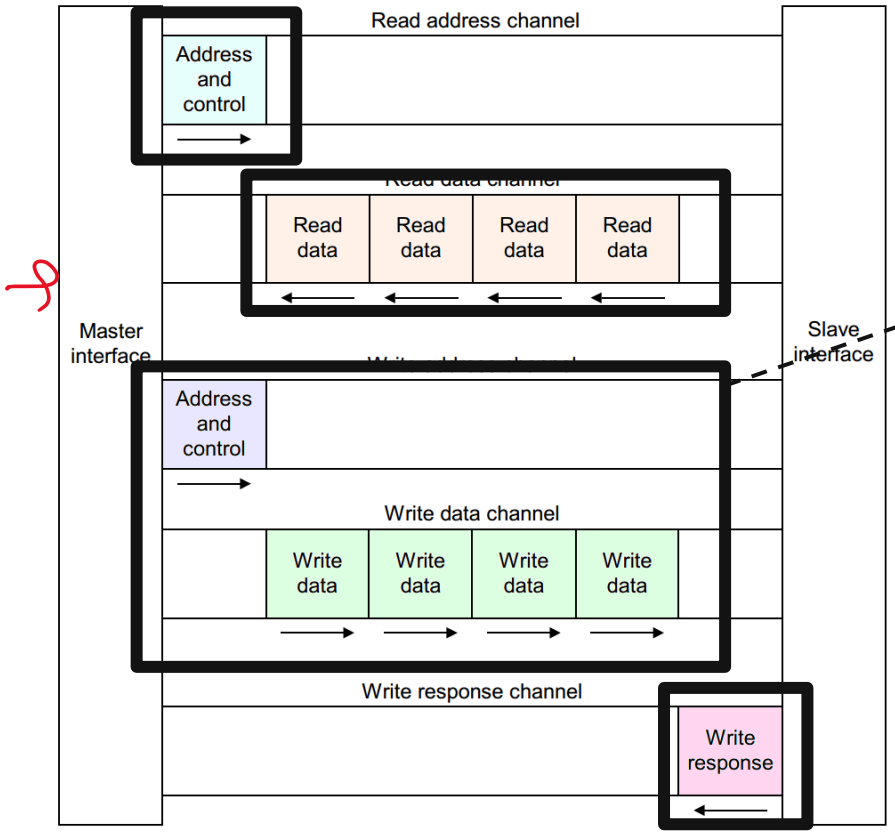
Packets/Packetization Concepts

- Define granularity of data transfer and applied switching technique
  - Phit (physical control digit) is a unit of data that is transferred on a link in a single cycle
  - Flit (flow control digit) is unit of switching
  - Typically, phitsize = flit size

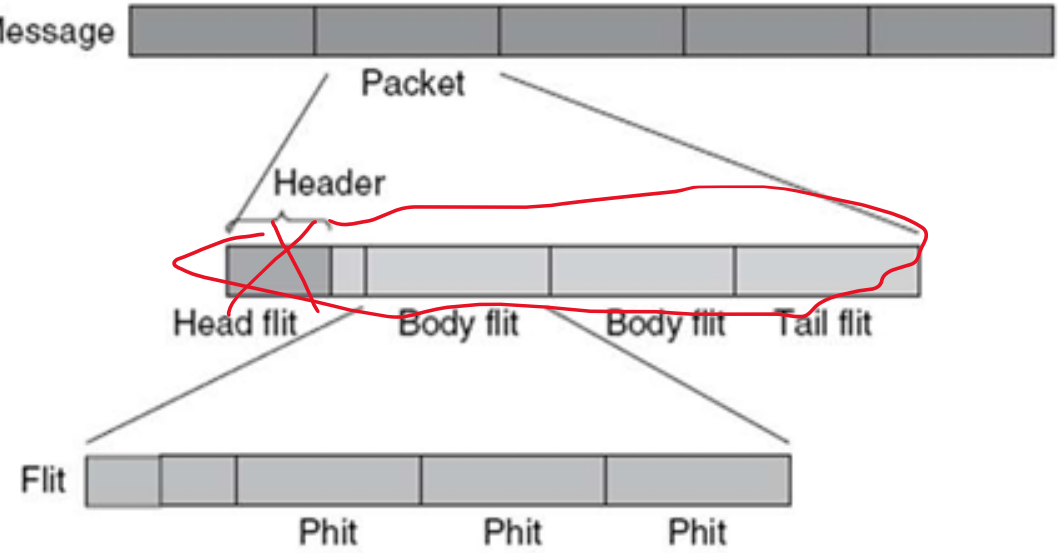


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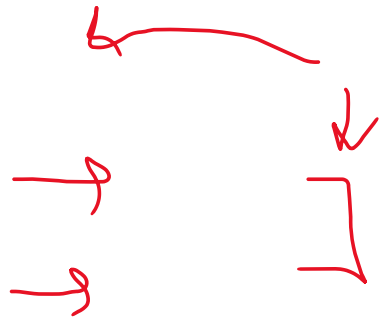
Packets/Packetization Concepts



Packed into one message.

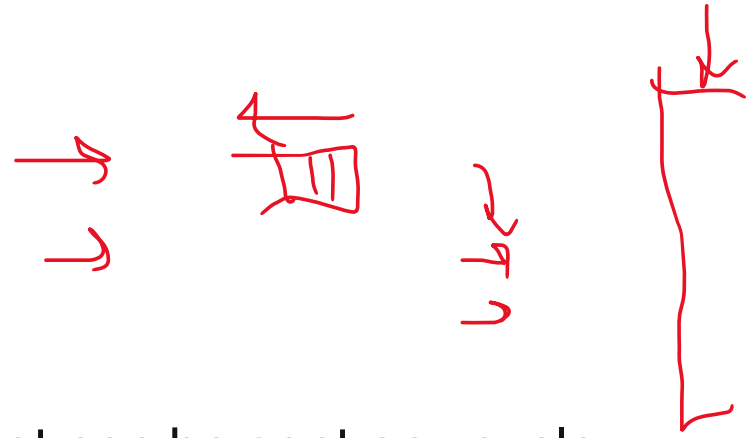


Flow Control

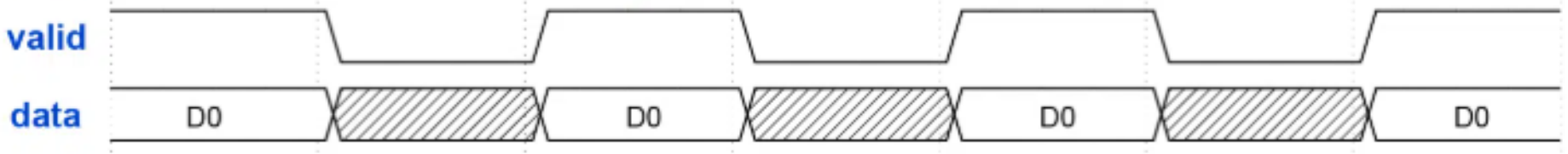


- Valid Only
- Valid/Ready
- Credit Based (Flit and Transaction Level)

Flow Control

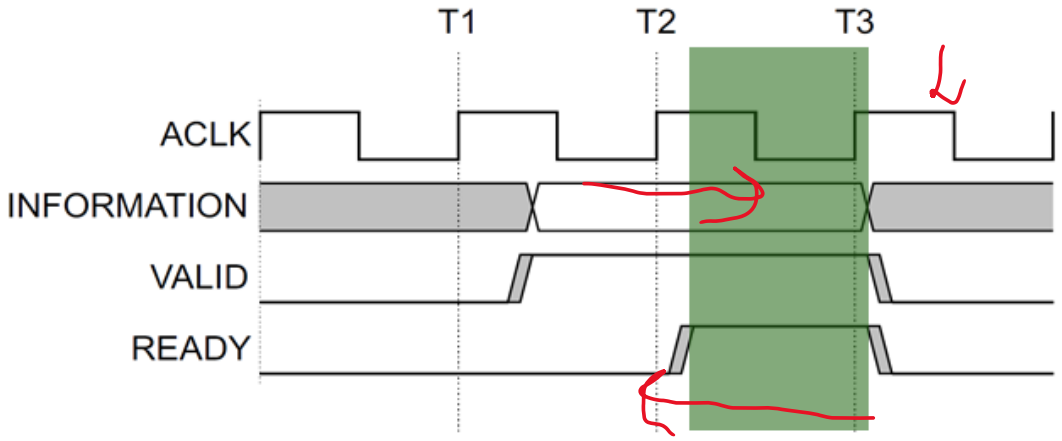


- Valid Only / No Backpressure
  - Downstream flows are designed to handle all transfers that can be sent any cycle.
  - For example, register accesses, any given register can be read or written in a single cycle.
  - There does need to be an indication that what on the line is valid.
  - Fairly rare outside some internal logic.



Flow Control

- Valid/Ready
  - Downstream will indicate any given cycle that it can accept a flit.
  - If ready is low a Valid/Ready interface can be retained or not retained.
    - Retained interfaces cannot change the transaction once valid is presented.
    - Non-Retained can change the transaction even after valid is presented (for example if a higher priority request comes in)



Flow Control

- Credit Based
  - Credits are units of transmission that an initiator can send knowing downstream can handle.
    - Could be in terms of flits, requests, types of requests, bytes of data.
  - Loaded / Propagated Credits
    - Credits can be loaded after reset so each initiator understands how much it can send imminently.
    - Credits can also be back propagated through a credit interface.
  - Static vs Dynamic Credits
    - Static credits means resources are partitioned at reset and wont change. Not very scalable.
    - Dynamic/Programmable credits mean that resources can be reallocated after reset. More scalable however protocol to change credits needs to be well defined or can deadlock.

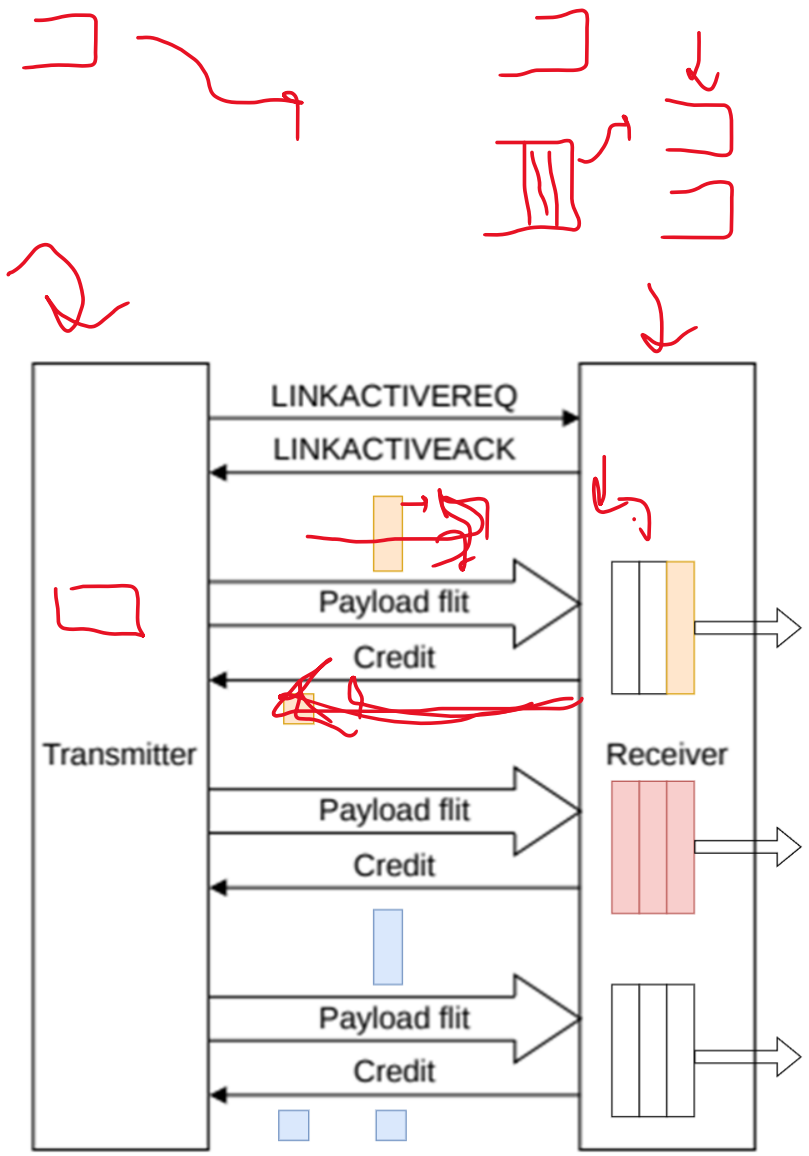
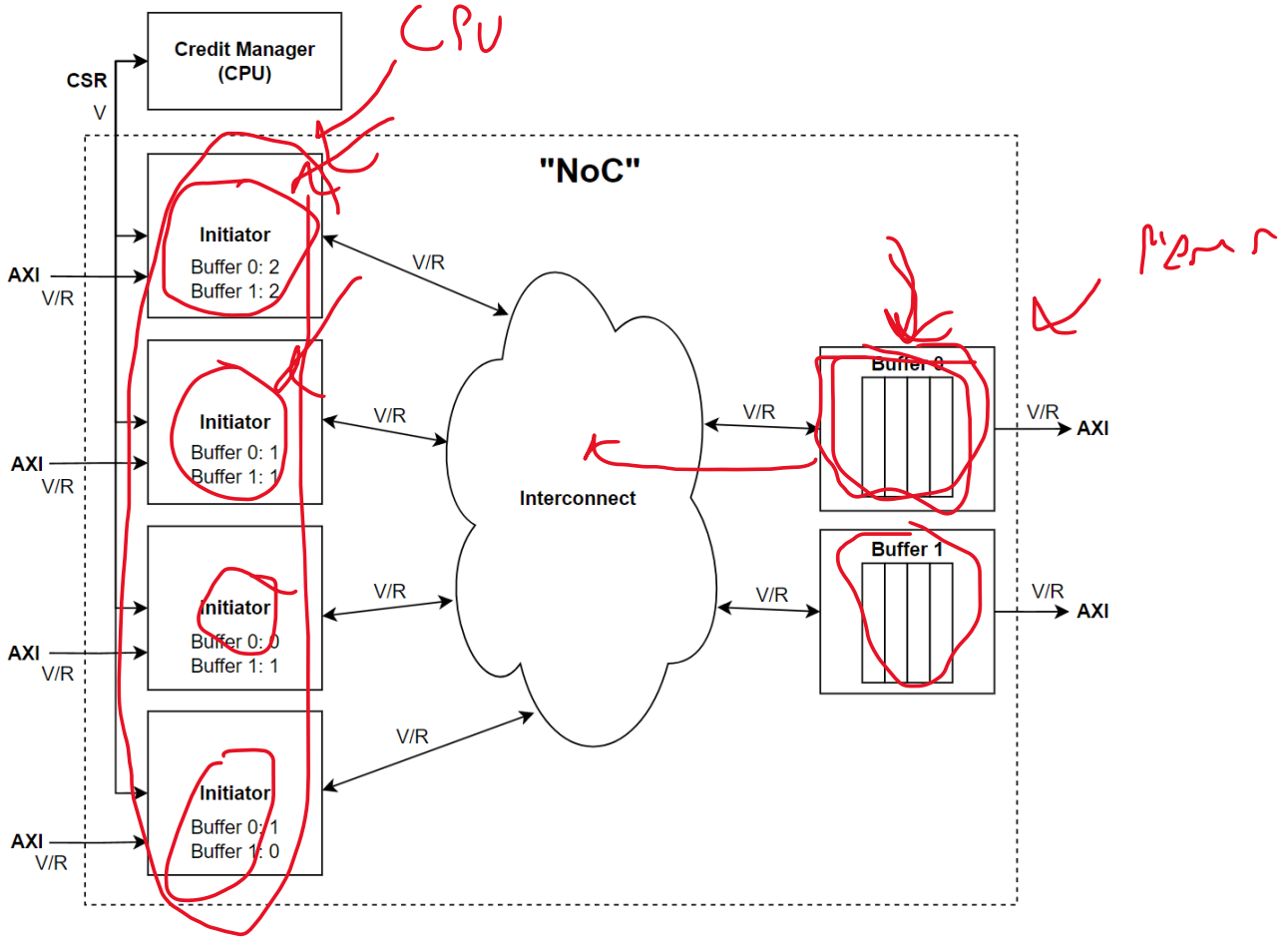


Figure B14.4: Example of a multiple channel unidirectional interface

Flow Control – Transaction Level

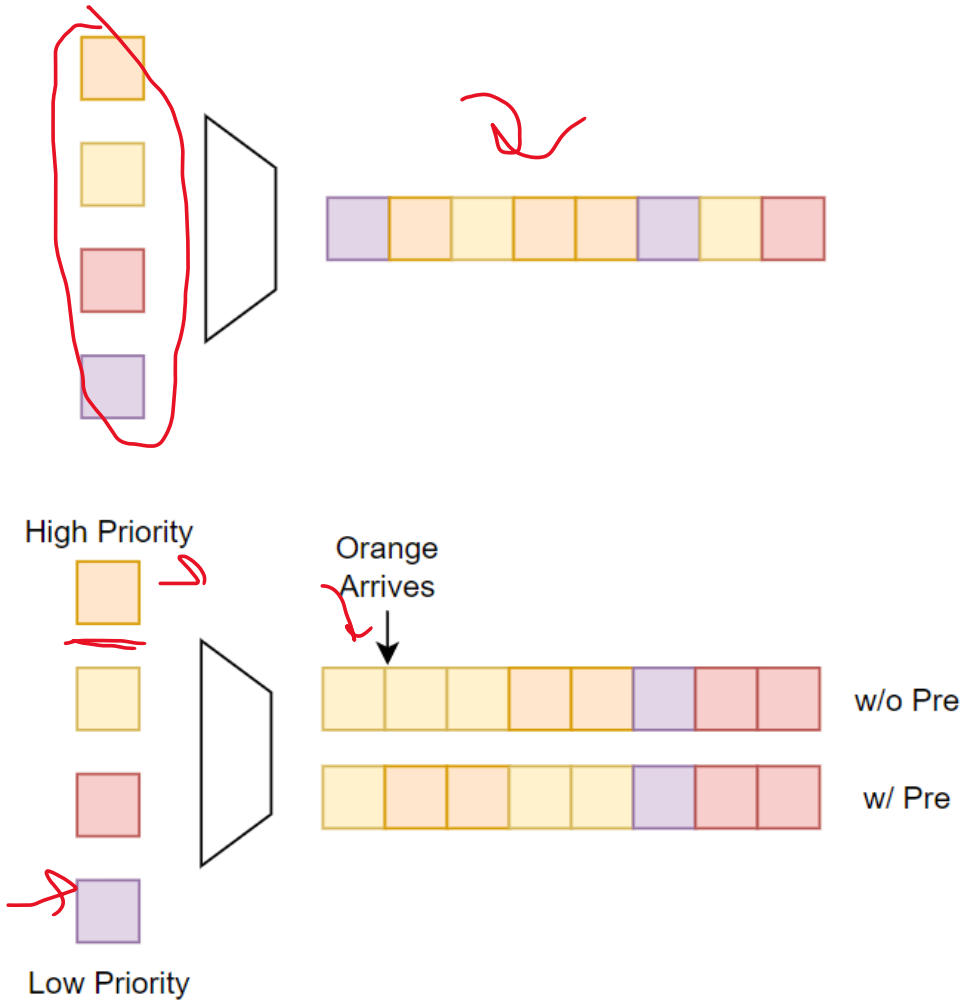


Arbitration Concepts

- Random
- Static priority
- Round Robin (RR)
- TDMA
- Dynamic
- Programmable

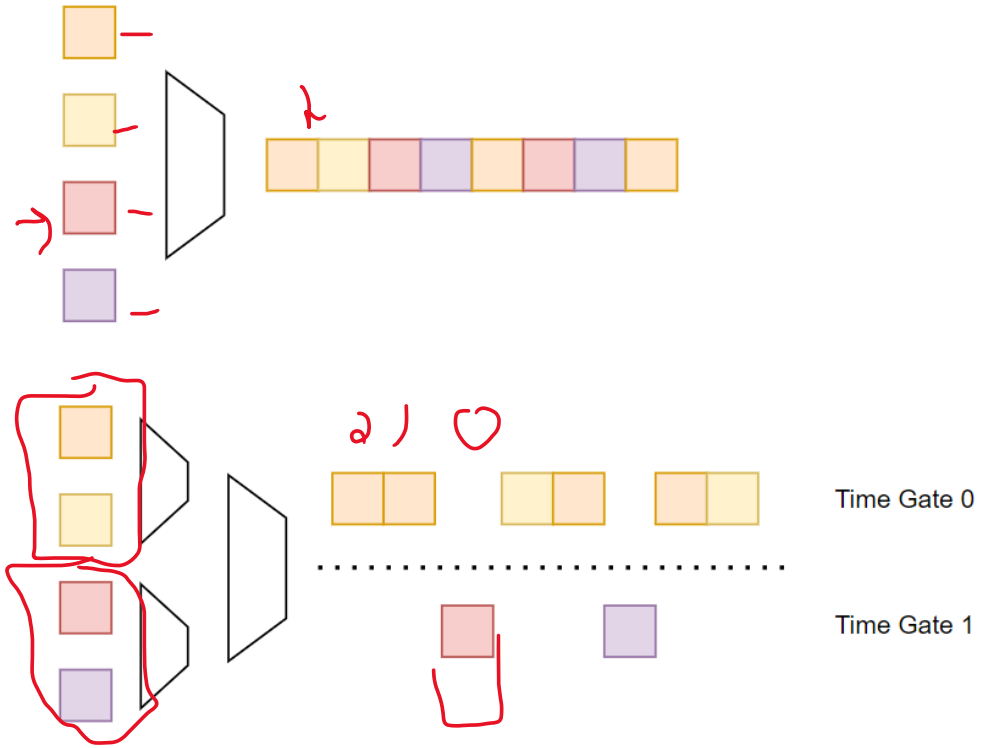
Arbitration Concepts

- Random/Statistical
  - Randomly select among the available flows. This randomness could be pure random or some statistical distribution.
- Static priority
  - At any time only select the highest priority flow available. A disadvantage is that it can be prone to deadlocks.
  - Can be preemptive or non preemptive.



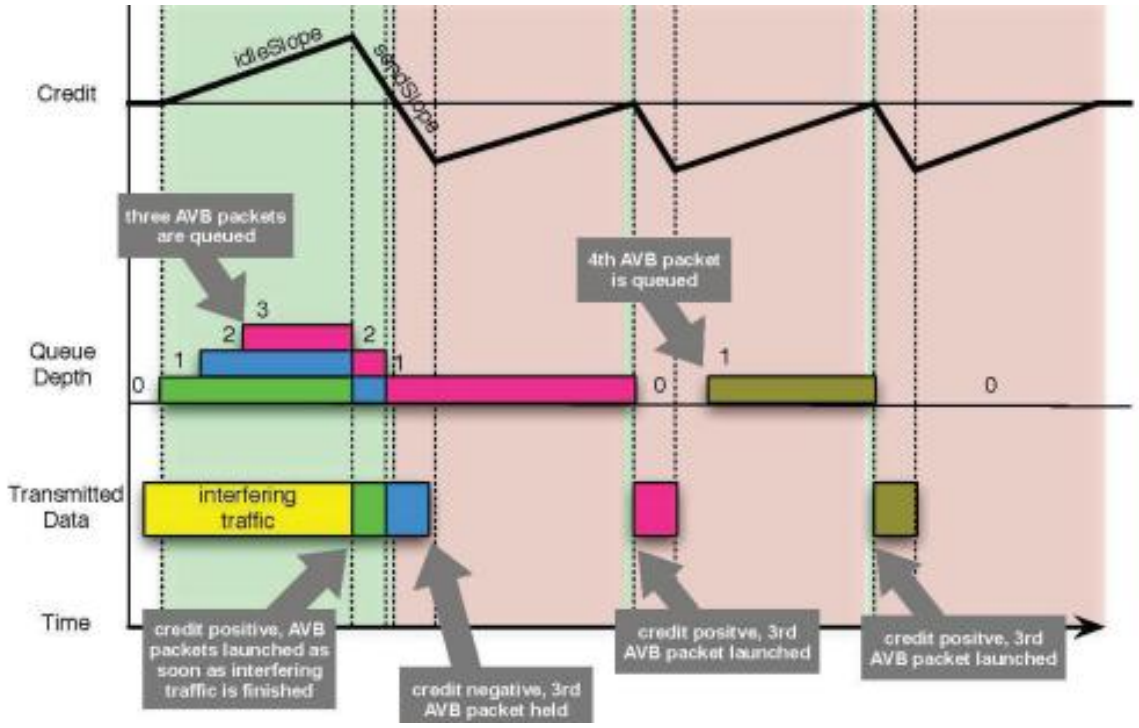
Arbitration Concepts

- Round Robin (RR)
  - Flows are selected in a rotating fashion.
  - Guarantees access to every flow.
  - However, may be difficult if different flows have different requirements. Can be high latency.
  - Work Conserving vs. Non-Work Conserving
- TDMA (Single and Two Level)
  - Each flow has a time window in which traffic can be sent.
  - Slots can be sized to bandwidth requirements.
  - With two level there can be multiple flows to choose per window allowing for higher utilization.



Arbitration Concepts

- Dynamic
  - Adapts to the traffic being sent.
  - For example, credit-based shaping of streaming traffic.
  - Need to keep track of some statistics or state to determine the balance of each flow.
- Programmable
  - Allows some outside statistical counter, or smarts to adjust priorities.
  - Can select schemes based on phases of traffic.



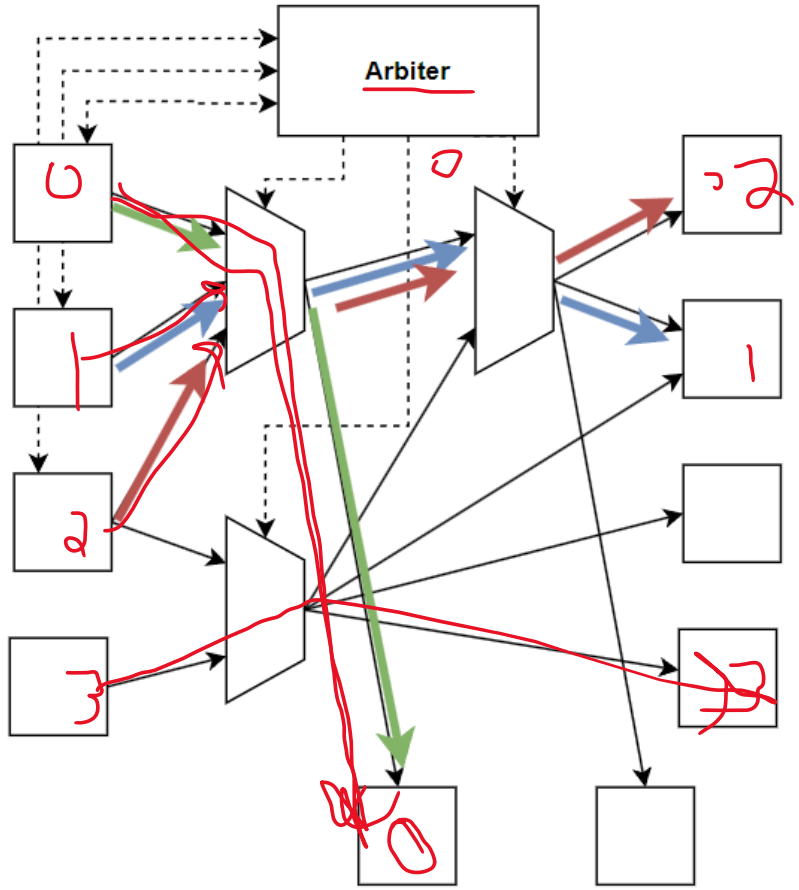
Microchip: TSN Application Note

Switching Concepts

- Circuit Switching
  - Bus-Like Design
- Packet-switched
  - Store-and-Forward
  - Virtual Cut-Through
  - Wormhole

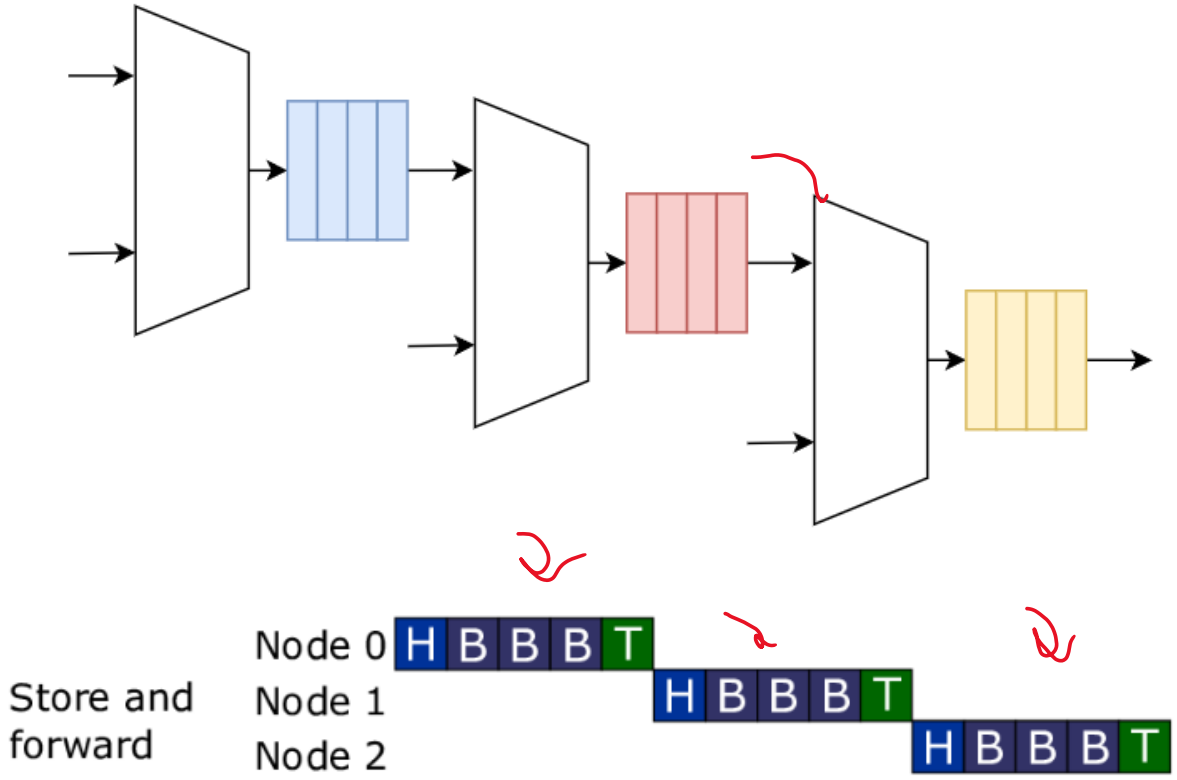
Switching Concepts

- Circuit Switching
  - The owner effectively reserves the physical resources (over multiple links) for transmission prior to sending the data.
  - Has very low latency transfers however does not scale wall.
- Virtual Circuit Switching
  - Multiple circuits to reserve per link.
  - Either time share a single buffer per link (lower bandwidth)
  - or have multiple buffers per link (more area)



Switching Concepts

- Store-and-Forward
  - Packet is sent from one router to the next only if the receiving router has buffer space for entire packet
  - Buffer size in the router is at least equal to the size of a packet
  - Disadvantage: excessive buffer requirements

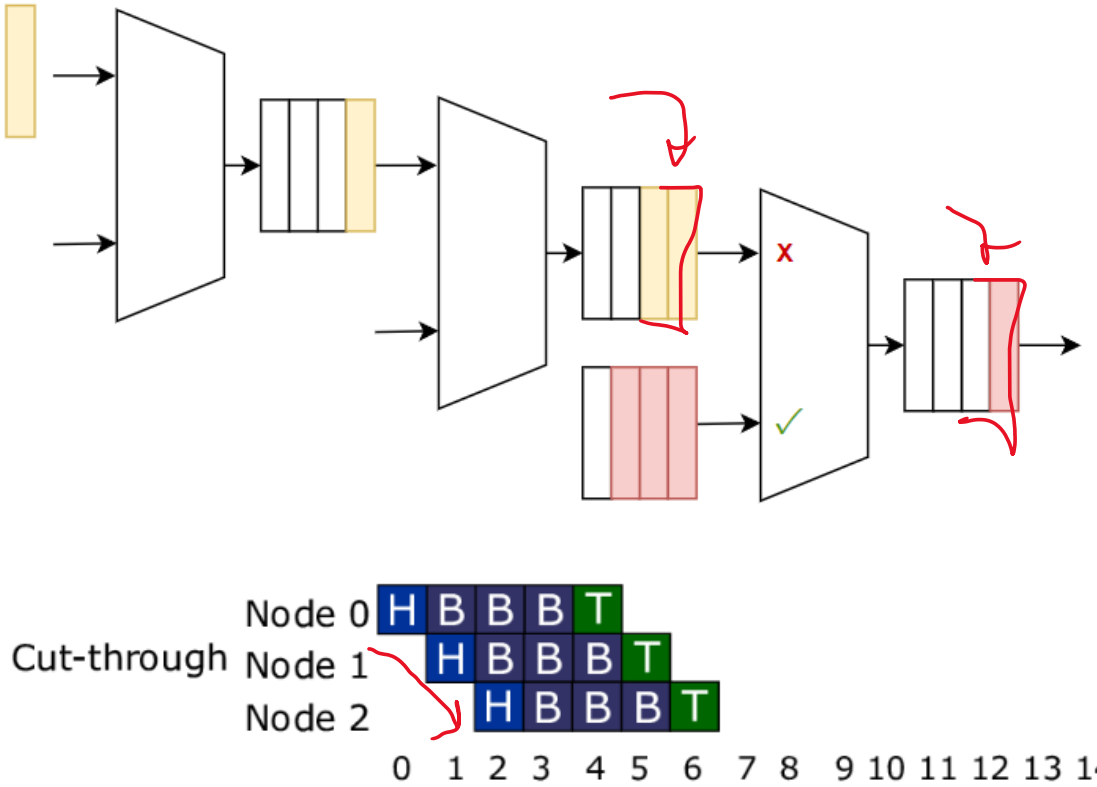


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Exploration of Manycore Systems based on Silicon Integrated Communication Networks

Switching Concepts

- Virtual Cut Through
  - Reduces router latency over SAF switching by forwarding first flit of a packet as soon as space for the entire packet is available in the next router
  - If no space is available in receiving buffer, no flits are sent, and the entire packet is buffered
  - Same buffering requirements as SAF switching

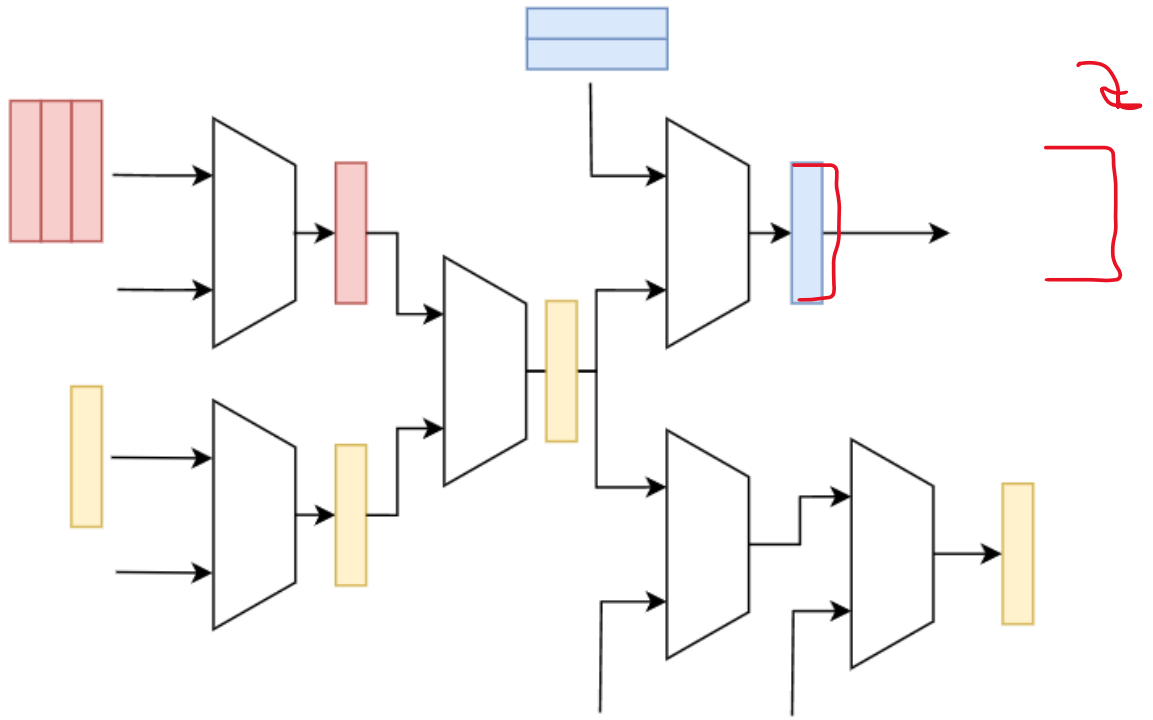


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Exploration of Manycore Systems based on Silicon Integrated Communication Networks

Switching Concepts

- Wormhole
  - Flit from a packet is forwarded to receiving router if space exists for that flit
  - Parts of the packet can be distributed among two or more routers
  - Buffer requirements are reduced to one flit, instead of an entire packet
  - More susceptible to deadlocks due to usage dependencies between links



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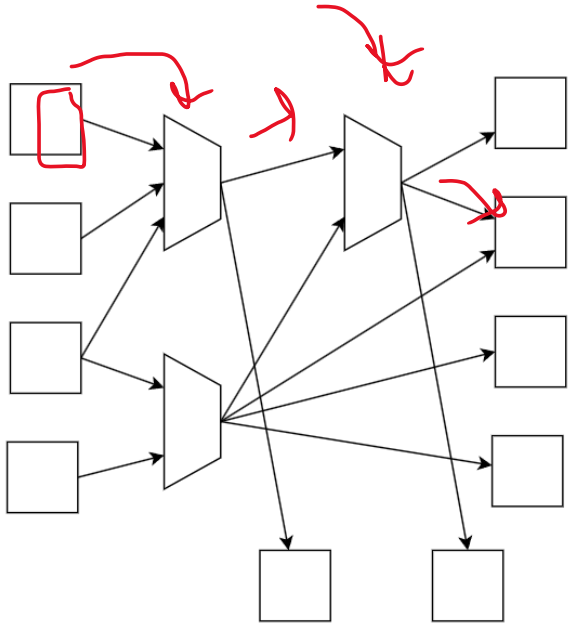
Routing Concepts

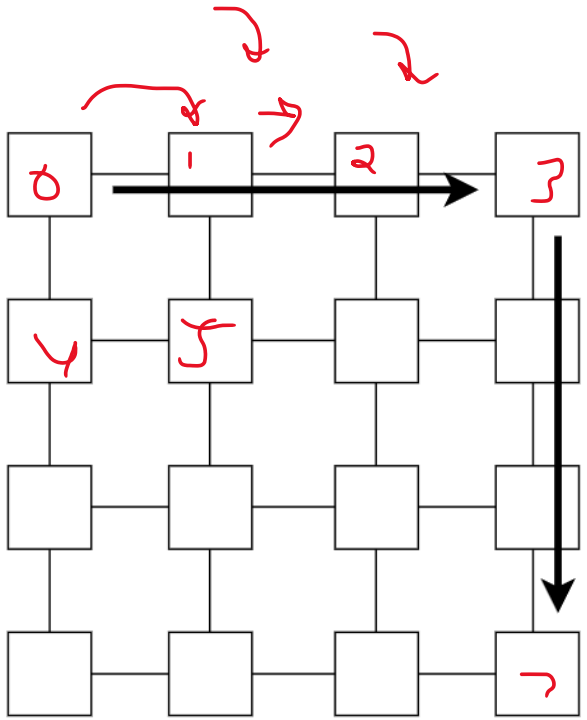
- Choices
  - Source vs. Destination Routing
  - Static vs. Dynamic Routing
  - Minimal vs. Non-Minimal Routing
- Example
  - XY Routing

- Considerations
  - Deadlocks
  - Livelocks
  - Fairness

Routing Concepts

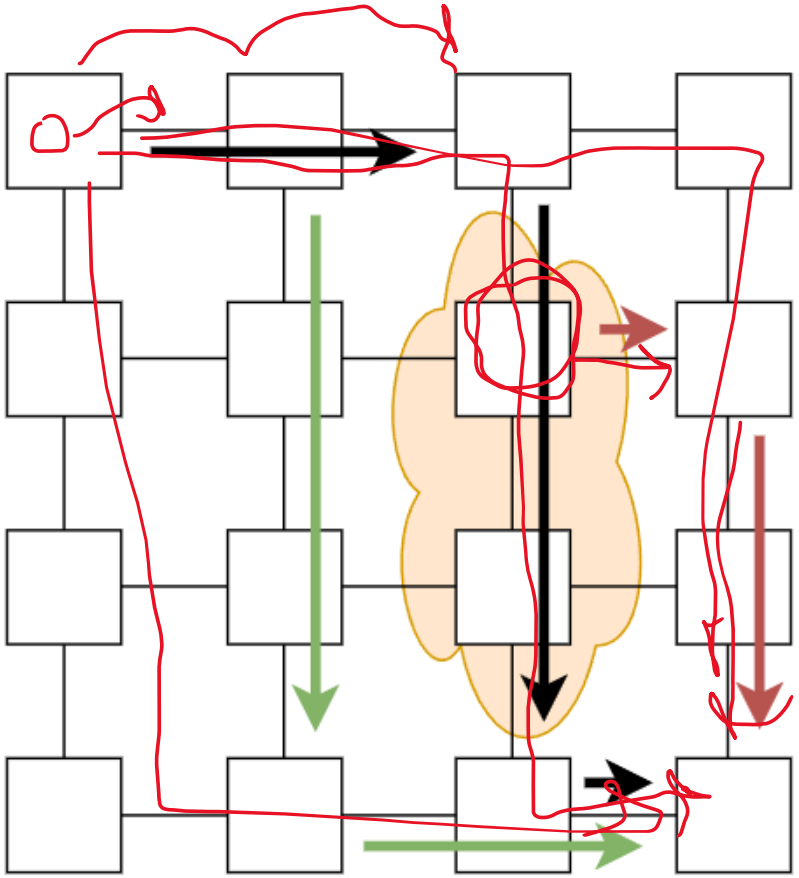
- Source vs. Destination Routing
  - In source routing the full route to the destination is known ahead of time (or additional alternative routes [expensive])
  - In destination routing each switch understands where in needs to make progress so the packet itself doesn't need to know the route. (through an address or a target id)





Routing Concepts

- Static vs. Dynamic Routing
  - Static (or deterministic) routing will move packets through the network on a set path and will not deviate.
    - There is also oblivious routing which can randomly or cyclically rotate through static routes
  - Dynamic (or adaptive) routing allows a packet to follow many routes responding to the current state of the network.



Routing Concepts

- Minimal vs. Non-Minimal
  - Minimal Routing means the packet will traverse along the shortest path.
  - Non-minimal the packet will not traverse the shortest path.
    - Many times, this is combined with adaptive routing to use a non-minimal path when minimal path is busy, or in fault tolerant schemes with a down link (in SoCs however this means the chip is most likely dead)
  - In more complex networks which routing is done more algorithmically many times proofs are done to show that some routing selection is “minimal”.

Routing Concepts

- Incorrect routing schemes can cause deadlock.

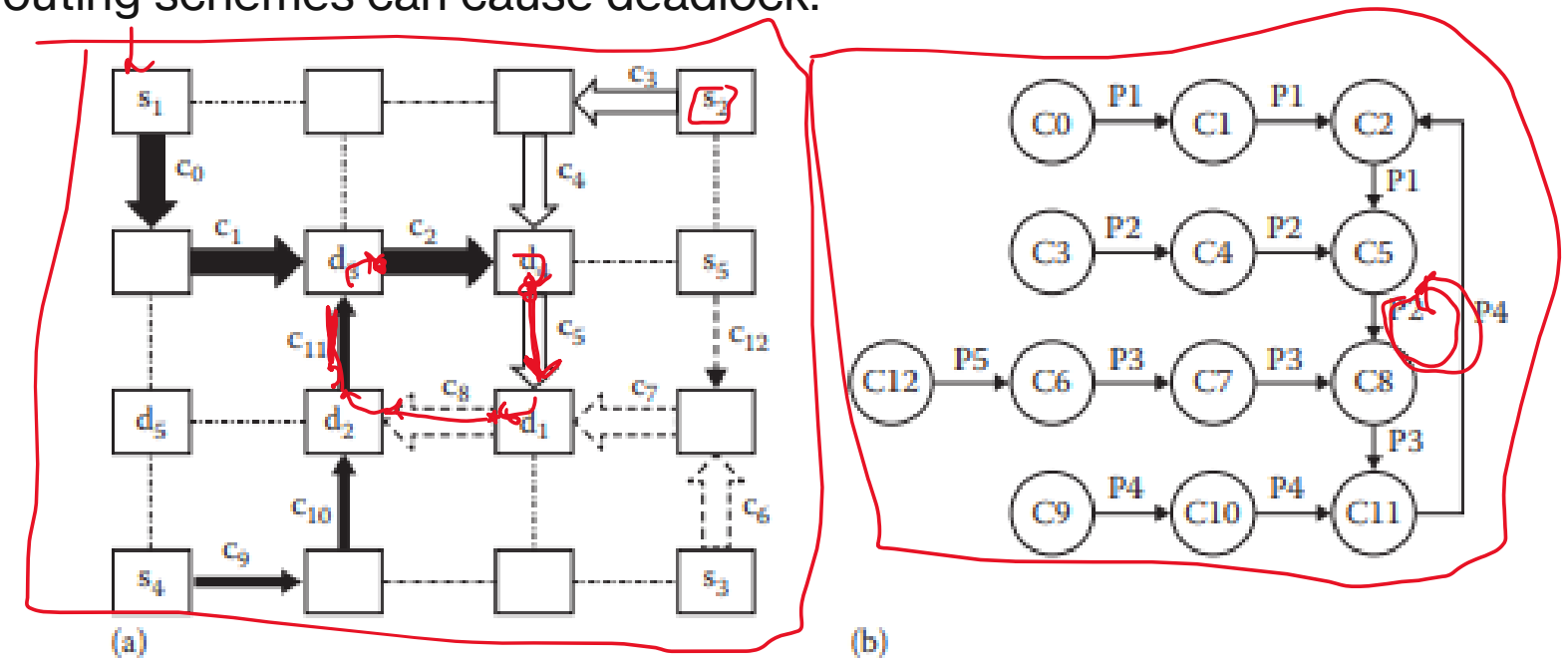


FIGURE 2.15
Routing-dependent deadlock in mesh network: (a) packet traversal path; (b) channel dependency graph.

Routing Concepts

- XY- Routing
  - Move first in the X direction to target node and then move in the Y direction.

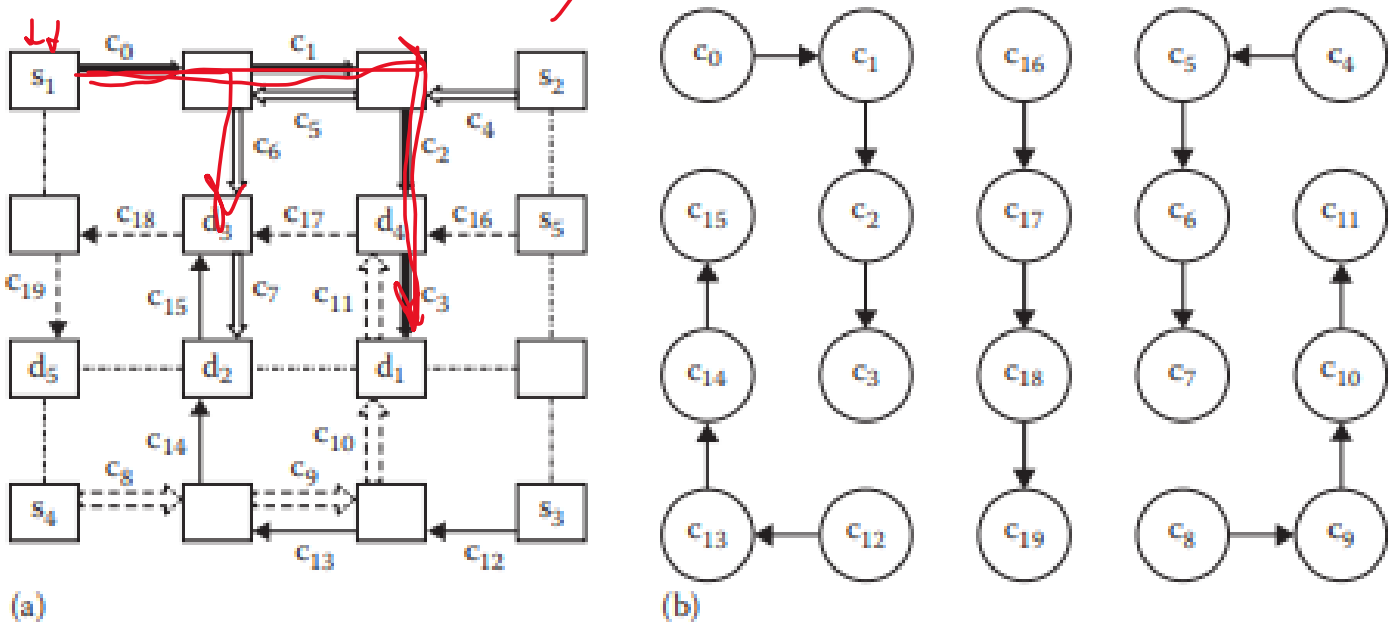


FIGURE 2.16
XY routing in 2D mesh topology (a) and its channel dependency graph (b).

Routing Concepts – Deadlock Free Rings

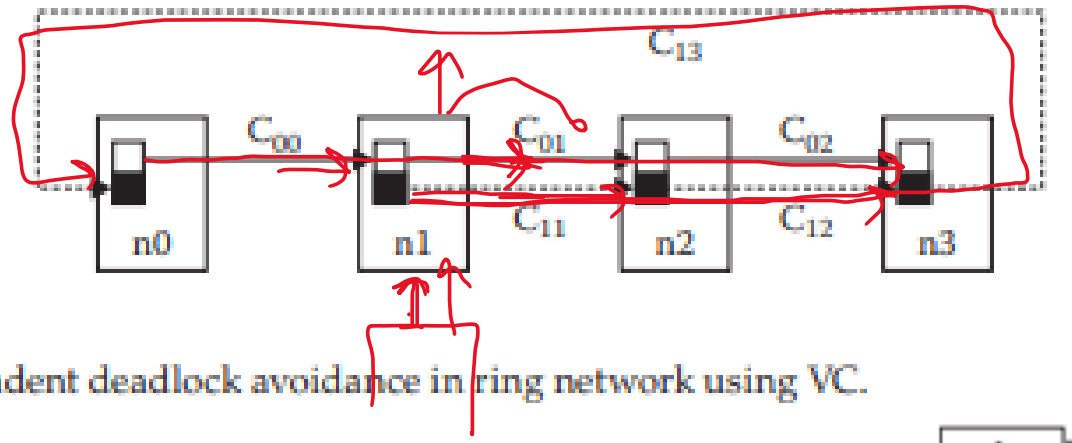
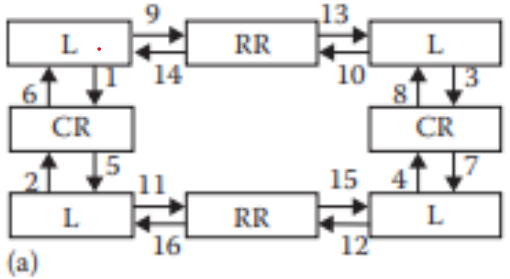
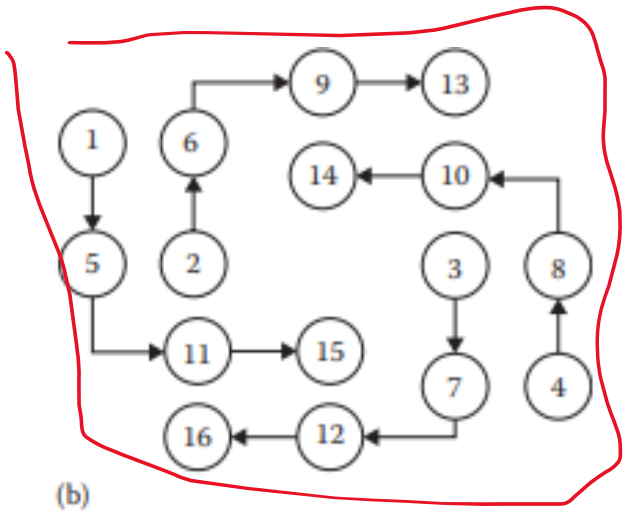


FIGURE 2.17
Routing-dependent deadlock avoidance in ring network using VC.



(a)



(b)

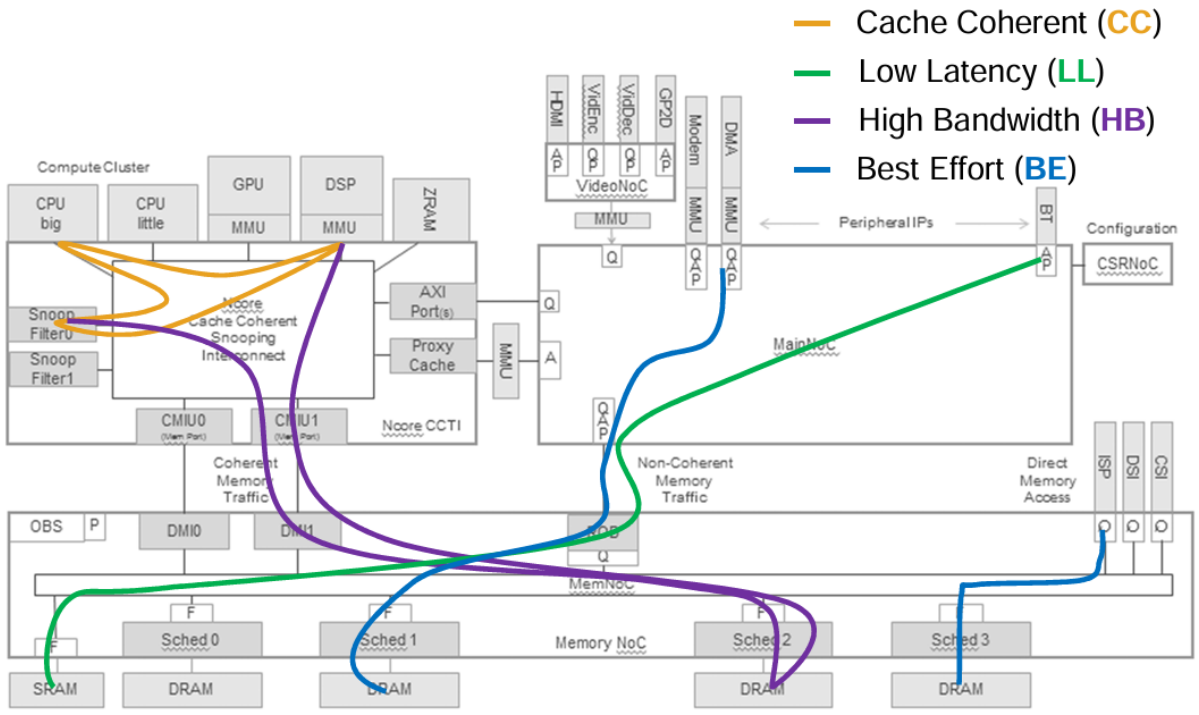
FIGURE 2.20
Labeling of 2 × 2 MoT (a) and its channel dependency graph (b).

QoS Concepts

- Traffic Classes
- Traffic Class Requirements
- Starvation

QoS Concepts - Example Schema

Memory Access Traffic Classes

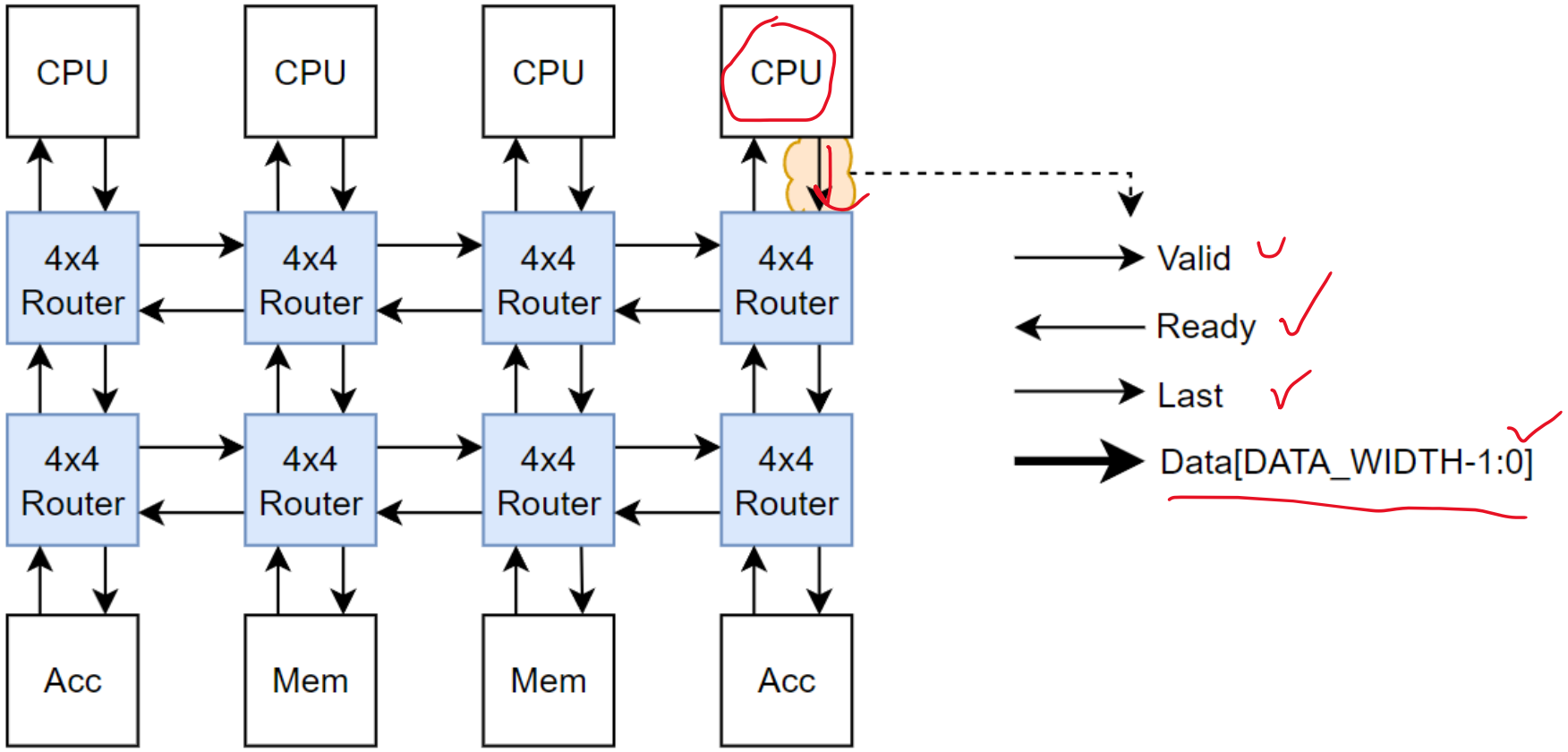


- **Cache Coherent** (**CC**) within Compute Cluster
- **Low Latency** (**LL**) to SRAM
- **High Bandwidth** (**HB**) to DRAM & Cache Fill
- **Best Effort** (**BE**) for Peripherals & DMA
- QoS for Video
- Multiple functional NoCs interacting
- Physically Constrained

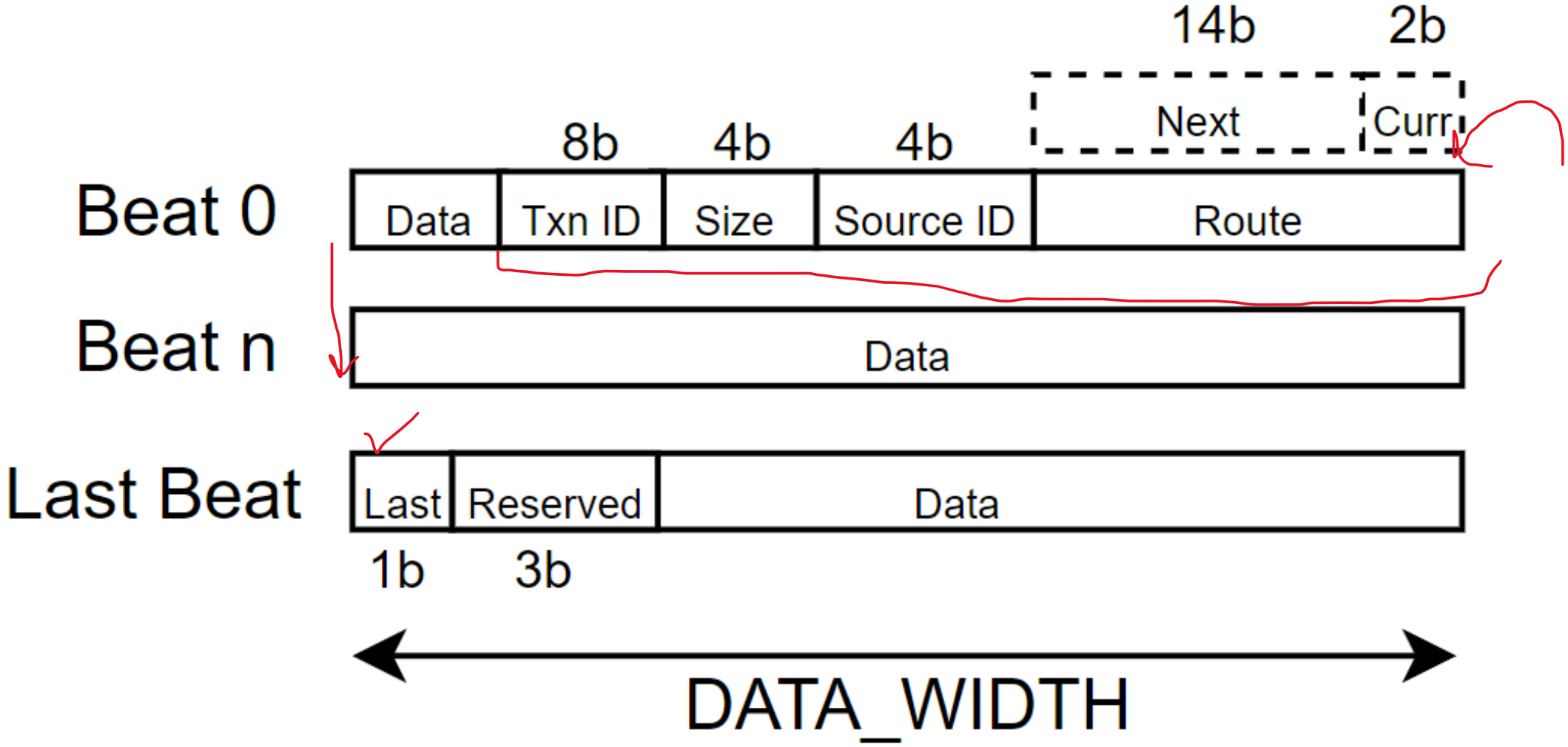
QoS Concepts

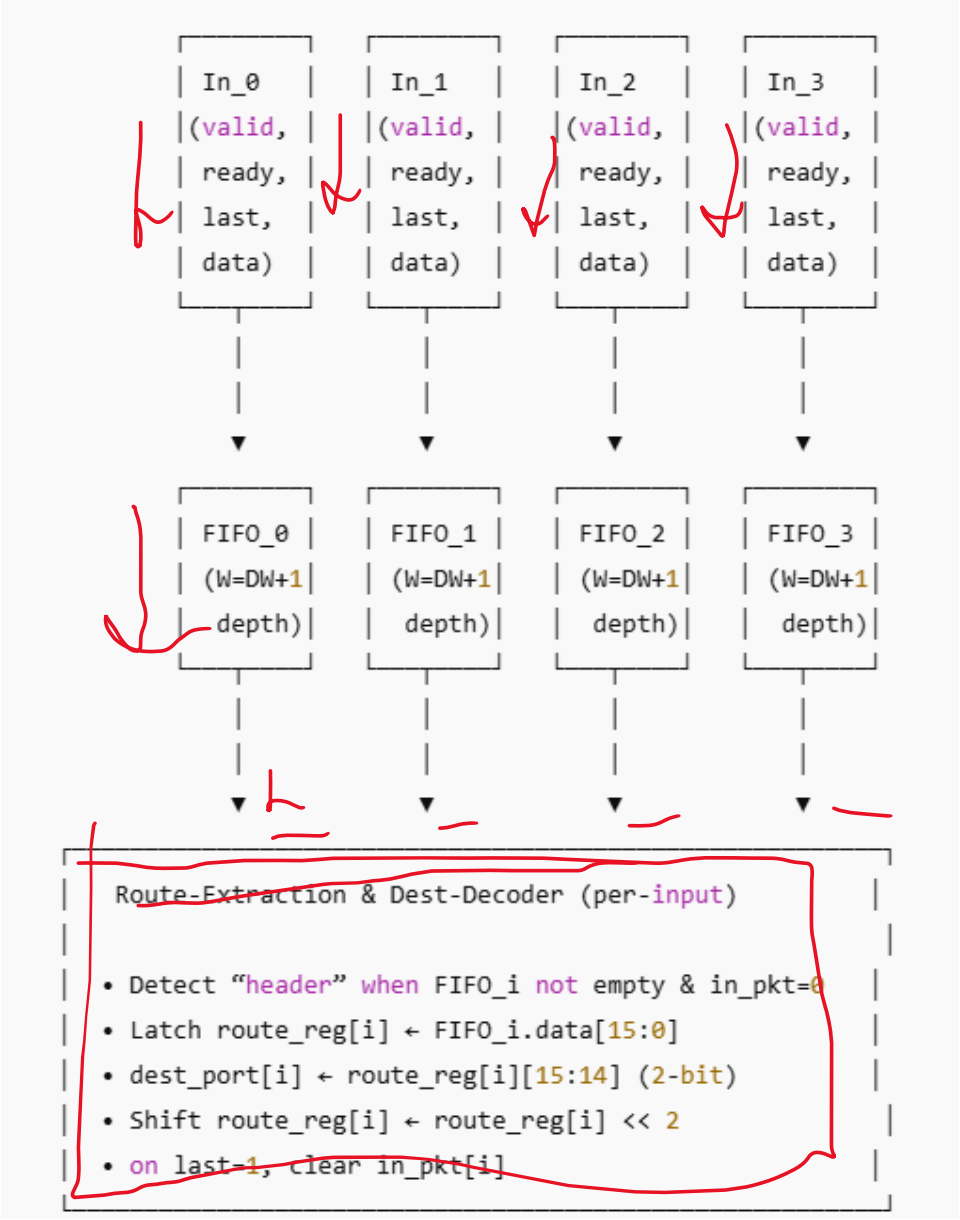
- Starvation
  - Any time QoS is used you need to have starvation mechanisms to detect and mitigate livelock scenarios.
  - Could either have arbitration force the lower traffic classes to have reserved BW (will take resources from higher BW classes)
  - Detect and mitigate where if transactions are stuck, stop accepting and then only allow progress on the starved transactions.
    - Don’t need to allocate resources for lower classes, however, can cause irregular behavior

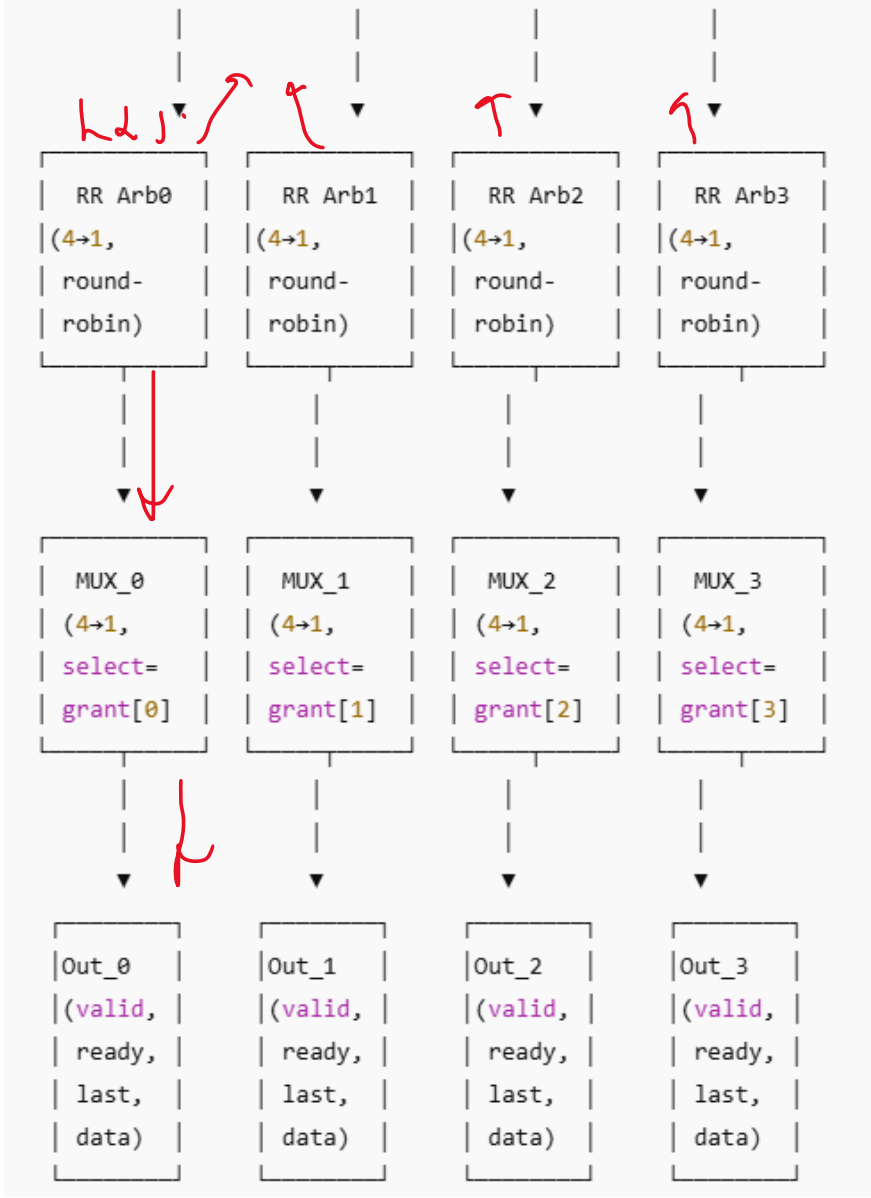
Switch/Router RTL Example



Switch/Router RTL Example




In_0 (valid, ready, last, data)
In_1 (valid, ready, last, data)
In_2 (valid, ready, last, data)
In_3 (valid, ready, last, data)
FIFO_0 (W=DW+1 depth)
FIFO_1 (W=DW+1 depth)
FIFO_2 (W=DW+1 depth)
FIFO_3 (W=DW+1 depth)
Route-Extraction & Dest-Decoder (per-input)
• Detect "header" when FIFO_i not empty & in_pkt=0
• Latch route_reg[i] ← FIFO_i.data[15:0]
• dest_port[i] ← route_reg[i][15:14] (2-bit)
• Shift route_reg[i] ← route_reg[i] << 2
• on last=1, clear in_pkt[i]


RR Arb0 (4→1, round-robin)
RR Arb1 (4→1, round-robin)
RR Arb2 (4→1, round-robin)
RR Arb3 (4→1, round-robin)
MUX_0 (4→1, select=grant[0])
MUX_1 (4→1, select=grant[1])
MUX_2 (4→1, select=grant[2])
MUX_3 (4→1, select=grant[3])
Out_0 (valid, ready, last, data)
Out_1 (valid, ready, last, data)
Out_2 (valid, ready, last, data)
Out_3 (valid, ready, last, data)

Backup (If Time)

- - Initiator / Target Logic
- - Multicast / Unicast
- - Fault Tolerant NoCs
- - Verification of Interconnects